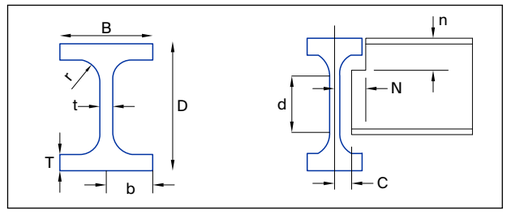
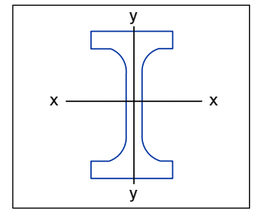
2.1
Parallel Flange I Sections (IPE)



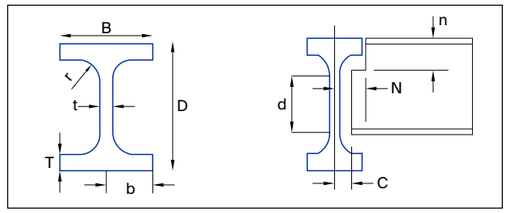
Section designation	Dimensions												
	Mass per Metre	Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
				Flange	Web				Flange	Web	End clearance	Notch	
		D	B	T	t	r	d	A	b/T	d/t	C	N	n
	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
IPE AA 80	4.9	78.0	46.0	4.2	3.2	5.0	59.6	6.3	3.90	18.6	4	32	10
IPE A 80	5.0	78.0	46.0	4.2	3.3	5.0	59.6	6.4	3.89	18.1	4	32	10
IPE 80	6.0	80.0	46.0	5.2	3.8	5.0	59.6	7.6	3.10	15.7	4	32	12
IPE AA 100	6.7	97.6	55.0	4.5	3.6	7.0	74.6	8.6	4.16	20.7	4	36	12
IPE A 100	6.9	98.0	55.0	4.7	3.6	7.0	74.6	8.8	3.98	20.7	4	36	12
IPE 100	8.1	100.0	55.0	5.7	4.1	7.0	74.6	10.3	3.24	18.2	4	36	14
IPE AA 120	8.4	117.0	64.0	4.8	3.8	7.0	93.4	10.7	4.81	24.6	4	42	12
IPE A 120	8.7	117.6	64.0	5.1	3.8	7.0	93.4	11.0	4.53	24.6	4	42	14
IPE 120	10.4	120.0	64.0	6.3	4.4	7.0	93.4	13.2	3.62	21.2	4	40	14
IPE AA 140	10.1	136.6	73.0	5.2	3.8	7.0	112.2	12.8	5.31	29.5	4	46	14
IPE A 140	10.5	137.4	73.0	5.6	3.8	7.0	112.2	13.4	4.93	29.5	4	46	14
IPE 140	12.9	140.0	73.0	6.9	4.7	7.0	112.2	16.4	3.93	23.9	4	46	14
IPE AA 160	12.3	156.4	82.0	5.6	4.0	9.0	131.2	15.7	5.36	32.8	4	50	14
IPE A 160	12.7	157.0	82.0	5.9	4.0	9.0	127.2	16.2	5.08	31.8	4	50	16
IPE 160	15.8	160.0	82.0	7.4	5.0	9.0	127.2	20.1	3.99	25.4	5	50	18
IPE AA 180	14.9	176.4	91.0	6.2	4.3	9.0	146.0	19.0	5.54	34.0	4	54	16
IPE A 180	15.4	177.0	91.0	6.5	4.3	9.0	146.0	19.6	5.28	34.0	4	54	16
IPE 180	18.8	180.0	91.0	8.0	5.3	9.0	146.0	23.9	4.23	27.5	5	54	18
IPE O 180	21.3	182.0	92.0	9.0	6.0	9.0	146.0	27.1	3.78	24.3	5	54	18
IPE AA 200	18.0	196.4	100.0	6.7	4.5	12.0	159.0	22.9	5.34	35.3	4	58	20
IPE A 200	18.4	197.0	100.0	7.0	4.5	12.0	159.0	23.5	5.11	35.3	4	58	20
IPE 200	22.4	200.0	100.0	8.5	5.6	12.0	159.0	28.5	4.14	28.4	5	58	22
IPE O 200	25.1	202.0	102.0	9.5	6.2	12.0	159.0	32.0	3.78	25.6	5	58	22
IPE AA 220	21.2	216.4	110.0	7.4	4.7	12.0	177.6	27.0	5.49	37.8	4	64	20
IPE A 220	22.2	217.0	110.0	7.7	5.0	12.0	177.6	28.3	5.26	35.5	5	64	20
IPE 220	26.2	220.0	110.0	9.2	5.9	12.0	177.6	33.4	4.35	30.1	5	64	22
IPE O 220	29.4	222.0	112.0	10.2	6.6	12.0	177.6	37.4	3.99	26.9	5	64	24
IPE AA 240	24.9	236.4	120.0	8.0	4.8	15.0	190.4	31.7	5.33	39.7	4	68	24

Dimensions	EN10365
Specification	EN10025
Size Range	IPE AA 80 to IPE 750 x 196

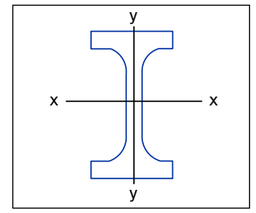


	Properties												
Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³	u	x	H dm⁶	J cm⁴
0.325	64.1	6.85	3.19	1.04	16.4	3	18.9	4.70	3.044	16.6	0.000090	0.38
0.325	64.4	6.85	3.18	1.04	16.5	3	19.0	4.70	3.025	16.5	0.000090	0.39
0.328	80.1	8.49	3.24	1.05	20.0	4	23.2	5.80	3.065	14.3	0.000120	0.67
0.396	136	12.6	3.98	1.21	27.9	5	31.9	7.20	3.279	18.7	0.000270	0.67
0.397	141	13.1	4.01	1.22	28.8	5	33.0	7.50	3.299	18.2	0.000280	0.73
0.400	171	15.9	4.07	1.24	34.2	6	39.4	9.20	3.333	15.8	0.000350	1.16
0.470	244	21.1	4.79	1.41	41.7	7	47.6	10.4	3.536	21.9	0.000660	0.89
0.472	257	22.4	4.83	1.42	43.8	7	49.9	11.0	3.577	21.2	0.000710	1.00
0.475	318	27.7	4.90	1.45	53.0	9	60.7	13.6	3.601	17.9	0.000890	1.69
0.546	407	33.8	5.64	1.63	59.7	9	67.6	14.5	3.818	24.8	0.00146	1.15
0.547	435	36.4	5.70	1.65	63.3	10	71.6	15.5	3.847	23.6	0.00158	1.34
0.551	541	44.9	5.74	1.65	77.3	12	88.3	19.3	3.854	19.6	0.00198	2.40
0.625	659	51.7	6.48	1.81	84.3	13	95.2	19.7	4.038	25.5	0.00293	1.75
0.619	689	54.4	6.53	1.83	87.8	13	99.1	20.7	4.062	24.7	0.00309	1.93
0.623	869	68.3	6.58	1.84	109	17	124	26.1	4.075	20.5	0.00396	3.54
0.693	1,020	78.1	7.32	2.03	116	17	131	26.7	4.282	26.9	0.00564	2.43
0.694	1,060	81.9	7.37	2.05	120	18	135	28.0	4.286	26.1	0.00593	2.67
0.698	1,320	101	7.42	2.05	146	22	166	34.6	4.299	21.8	0.00743	4.73
0.705	1,500	117	7.45	2.08	165	26	189	39.9	4.321	19.7	0.00874	6.65
0.763	1,530	112	8.19	2.21	156	22	176	35.0	4.472	26.3	0.0101	3.81
0.764	1,590	117	8.23	2.23	162	23	182	36.5	4.496	25.5	0.0105	4.14
0.768	1,940	142	8.26	2.24	194	29	221	44.6	4.492	22.0	0.0130	6.92
0.779	2,210	169	8.32	2.30	219	33	249	51.9	4.547	20.1	0.0156	9.36
0.843	2,220	165	9.07	2.47	205	30	230	46.5	4.727	27.2	0.0179	5.06
0.843	2,320	171	9.05	2.46	214	31	240	48.5	4.709	26.4	0.0187	5.68
0.848	2,770	205	9.11	2.48	252	37	285	58.1	4.725	22.9	0.0227	9.03
0.858	3,130	240	9.16	2.53	282	43	321	66.9	4.777	21.0	0.0268	12.2
0.917	3,150	231	9.97	2.70	267	39	298	60.0	4.949	26.4	0.0301	7.61

2.1
Parallel Flange I Sections (IPE)

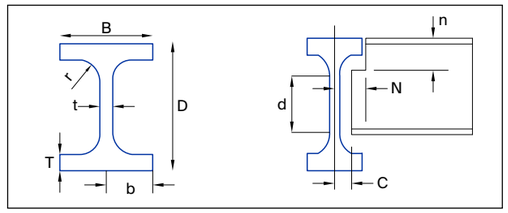


Dimensions	EN10365
Specification	EN10025
Size Range	IPE AA 80 to IPE 750 x 196

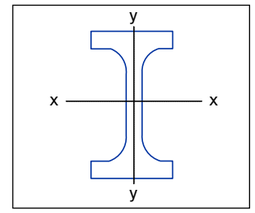


Section designation	Dimensions													Properties												
	Mass per Metre	Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing			Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
				Flange	Web				Flange	Web	End clearance	Notch														
		D	B	T	t	r	d	A	b/T	d/t	C	N	n	Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	u	x	H	J
	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm	m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
IPE A 240	26.2	237.0	120.0	8.3	5.2	15.0	190.4	33.3	5.11	36.6	5	68	24	0.918	3,290	240	9.94	2.68	278	40	312	62.4	4.936	25.6	0.0313	8.50
IPE 240	30.7	240.0	120.0	9.8	6.2	15.0	190.4	39.1	4.28	30.7	5	68	26	0.922	3,890	284	9.97	2.69	324	47	367	73.9	4.941	22.6	0.0374	13.0
IPE O 240	34.3	242.0	122.0	10.8	7.0	15.0	190.4	43.7	3.94	27.2	6	68	26	0.932	4,370	329	10.0	2.74	361	54	410	84.4	4.976	20.8	0.0437	17.1
IPE A 270	30.7	267.0	135.0	8.7	5.5	15.0	219.6	39.2	5.72	39.9	5	76	24	1.04	4,920	358	11.2	3.02	368	53	413	82.3	5.232	28.3	0.0595	10.4
IPE 270	36.1	270.0	135.0	10.2	6.6	15.0	219.6	45.9	4.82	33.3	5	76	26	1.04	5,790	420	11.2	3.02	429	62	484	97.0	5.230	24.9	0.0706	15.9
IPE O 270	42.3	274.0	136.0	12.2	7.5	15.0	219.6	53.8	4.04	29.3	6	76	28	1.05	6,950	514	11.4	3.09	507	76	575	118	5.289	21.7	0.0876	25.0
IPE A 300	36.5	297.0	150.0	9.2	6.1	15.0	248.6	46.5	6.19	40.8	5	82	26	1.16	7,170	519	12.4	3.34	483	69	542	107	5.495	30.3	0.107	13.3
IPE 300	42.2	300.0	150.0	10.7	7.1	15.0	248.6	53.8	5.28	35.0	6	82	26	1.16	8,360	604	12.5	3.35	557	81	628	125	5.498	26.8	0.126	19.9
IPE O 300	49.3	304.0	152.0	12.7	8.0	15.0	248.6	62.8	4.49	31.1	6	82	28	1.17	9,990	746	12.6	3.45	658	98	744	153	5.580	23.4	0.158	31.0
IPE A 330	43.0	327.0	160.0	10.0	6.5	18.0	271.0	54.7	5.88	41.7	5	88	28	1.25	10,200	685	13.7	3.54	626	86	702	133	5.664	29.9	0.172	19.6
IPE 330	49.1	330.0	160.0	11.5	7.5	18.0	271.0	62.6	5.07	36.1	6	88	30	1.25	11,800	788	13.7	3.55	713	99	804	154	5.666	26.8	0.199	28.1
IPE O 330	57.0	334.0	162.0	13.5	8.5	18.0	271.0	72.6	4.35	31.9	6	88	32	1.27	13,900	960	13.8	3.64	833	119	943	185	5.738	23.7	0.246	42.2
IPE A 360	50.2	357.6	170.0	11.5	6.6	18.0	298.6	64.0	5.54	45.2	5	92	30	1.35	14,500	944	15.1	3.84	812	111	907	172	5.911	29.9	0.282	27.4
IPE 360	57.1	360.0	170.0	12.7	8.0	18.0	298.6	72.7	4.96	37.3	6	92	32	1.35	16,300	1,040	15.0	3.79	904	123	1,020	191	5.859	27.3	0.314	37.4
IPE O 360	66.0	364.0	172.0	14.7	9.2	18.0	298.6	84.1	4.31	32.5	7	92	34	1.37	19,000	1,250	15.1	3.86	1,050	146	1,190	227	5.910	24.2	0.380	55.7
IPE A 400	57.4	397.0	180.0	12.0	7.0	21.0	331.0	73.1	5.46	47.3	6	98	34	1.46	20,300	1,170	16.7	4.00	1,020	130	1,140	202	6.041	30.9	0.432	36.2
IPE 400	66.3	400.0	180.0	13.5	8.6	21.0	331.0	84.5	4.79	38.5	6	96	36	1.47	23,100	1,320	16.6	3.95	1,160	146	1,310	229	5.988	28.0	0.490	51.3
IPE O 400	75.7	404.0	182.0	15.5	9.7	21.0	331.0	96.4	4.20	34.1	7	98	38	1.48	26,800	1,560	16.7	4.03	1,320	172	1,500	269	6.046	25.1	0.588	73.3
IPE V 400	84.0	408.0	182.0	17.5	10.6	21.0	331.0	107	3.70	31.2	7	96	40	1.49	30,100	1,770	16.8	4.06	1,480	194	1,680	304	6.074	22.9	0.673	99.6
IPE A 450	67.2	447.0	190.0	13.1	7.6	21.0	378.8	85.6	5.36	49.8	6	102	36	1.60	29,800	1,500	18.7	4.19	1,330	158	1,490	246	6.181	33.0	0.705	47.1
IPE 450	77.6	450.0	190.0	14.6	9.4	21.0	378.8	98.8	4.75	40.3	7	102	36	1.61	33,700	1,680	18.5	4.12	1,500	176	1,700	276	6.118	29.9	0.791	66.7
IPE O 450	92.4	456.0	192.0	17.6	11.0	21.0	378.8	118	3.95	34.4	8	102	40	1.62	40,900	2,080	18.7	4.21	1,800	217	2,050	341	6.183	25.7	0.998	109
IPE V 450	107	460.0	194.0	19.6	12.4	21.0	378.8	132	3.56	30.5	8	102	42	1.64	46,200	2,400	18.7	4.26	2,010	247	2,300	389	6.218	23.5	1.16	149
IPE A 500	79.4	497.0	200.0	14.5	8.4	21.0	426.0	101	5.16	50.7	6	106	36	1.74	42,900	1,940	20.6	4.38	1,730	194	1,950	302	6.322	34.2	1.12	64.3
IPE 500	90.7	500.0	200.0	16.0	10.2	21.0	426.0	116	4.62	41.8	7	106	38	1.74	48,200	2,140	20.4	4.31	1,930	214	2,190	336	6.257	31.1	1.25	89.1
IPE O 500	107	506.0	202.0	19.0	12.0	21.0	426.0	137	3.89	35.5	8	106	40	1.76	57,800	2,620	20.6	4.38	2,280	260	2,610	409	6.307	26.9	1.55	143
IPE V 500	129	514.0	204.0	23.0	14.2	21.0	426.0	164	3.21	30.0	9	106	44	1.78	70,700	3,270	20.7	4.46	2,750	320	3,170	506	6.368	22.9	1.97	242
IPE A 550	92.1	547.0	210.0	15.7	9.0	24.0	467.6	117	4.87	52.0	7	112	40	1.87	60,000	2,430	22.6	4.55	2,190	232	2,480	362	6.451	34.4	1.71	89.3

2.1
Parallel Flange I Sections (IPE)

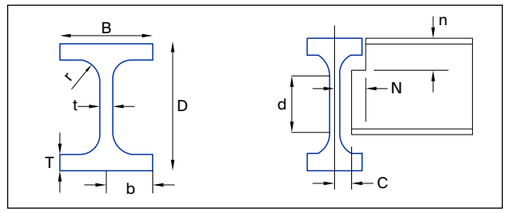


Dimensions	EN10365
Specification	EN10025
Size Range	IPE AA 80 to IPE 750 x 196



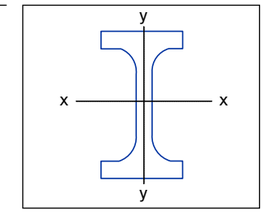
Section designation	Dimensions													Properties											
	Mass per Metre	Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing			Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
				Flange	Web				Flange	Web	End clearance	Notch														
		D	B	T	t	r	d	A	b/T	d/t	C	N	n	Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	u	x	H	J
	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm	m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
IPE 550	106	550.0	210.0	17.2	11.1	24.0	467.6	134	4.39	42.1	8	110	42	1.88	67,100	2,670	22.4	4.45	2,440	254	2,790	401	6.365	31.4	1.88	123
IPE O 550	123	556.0	212.0	20.2	12.7	24.0	467.6	156	3.75	36.8	8	110	46	1.89	79,200	3,220	22.5	4.55	2,850	304	3,260	481	6.428	27.6	2.30	187
IPE V 550	159	566.0	216.0	25.2	17.1	24.0	467.6	202	2.99	27.3	11	110	50	1.92	102,000	4,260	22.5	4.59	3,620	395	4,200	632	6.449	22.5	3.12	372
IPE A 600	108	597.0	220.0	17.5	9.8	24.0	514.0	137	4.63	52.4	7	116	42	2.01	82,900	3,120	24.6	4.77	2,780	283	3,140	442	6.603	34.7	2.61	122
IPE 600	122	600.0	220.0	19.0	12.0	24.0	514.0	156	4.21	42.8	8	114	44	2.01	92,100	3,390	24.3	4.66	3,070	308	3,510	486	6.510	31.9	2.85	165
IPE O 600	154	610.0	224.0	24.0	15.0	24.0	514.0	197	3.35	34.3	10	116	48	2.04	118,000	4,520	24.5	4.79	3,880	404	4,470	640	6.600	26.1	3.86	316
IPE V 600	184	618.0	228.0	28.0	18.0	24.0	514.0	234	2.89	28.6	11	116	52	2.07	142,000	5,570	24.6	4.88	4,580	488	5,320	780	6.653	22.7	4.85	506
IPE 750 x 134	134	750.0	264.0	15.5	12.0	17.0	685.0	171	7.03	57.1	8	136	34	2.50	151,000	4,790	29.7	5.30	4,020	362	4,640	570	6.910	49.5	6.44	120
IPE 750 x 147	147	753.0	265.0	17.0	13.2	17.0	685.0	188	6.41	51.9	9	136	34	2.51	166,000	5,290	29.8	5.31	4,410	399	5,110	631	6.917	45.4	7.14	157
IPE 750 x 173	173	762.0	267.0	21.6	14.4	17.0	685.0	221	5.06	47.6	9	138	40	2.53	206,000	6,870	30.5	5.57	5,400	515	6,220	810	7.105	37.8	9.39	270
IPE 750 x 196	196	770.0	268.0	25.4	15.6	17.0	685.0	251	4.30	43.9	10	138	44	2.55	240,000	8,180	31.0	5.71	6,240	610	7,170	959	7.201	33.0	11.3	406
IPE 750 x 220	220	779.0	266.0	30.0	16.5	17.0	685.1	281	3.59	41.5	10	136	48	2.56	278,000	9,440	31.5	5.80	7,140	710	8,200	1,110	7.272	28.8	13.2	607

2.2
Wide Flange Beams (HE)



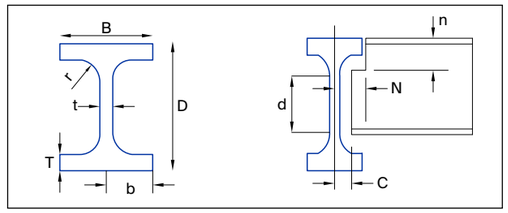
Section designation	Dimensions												
	Mass per Metre	Depth of Section	Width of Section	Thickness		Root	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
				Flange	Web				Flange	Web	End clearance	Notch	
		D	B	T	t	r	d	A	b/T	d/t	C	N	n
	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
HE 100 AA	12.2	91.0	100.0	5.5	4.2	12.0	56.0	15.6	13.3	6.53	4	58	18
HE 100 A	16.7	96.0	100.0	8.0	5.0	12.0	56.0	21.2	11.2	4.44	5	58	20
HE 100 B	20.4	100.0	100.0	10.0	6.0	12.0	56.0	26.0	9.33	3.50	5	58	22
HE 100 C	30.9	110.0	103.0	15.0	9.0	12.0	56.0	39.3	6.22	2.33	7	58	28
HE 100 M	41.8	120.0	106.0	20.0	12.0	12.0	56.0	53.2	4.67	1.75	8	58	32
HE 120 AA	14.6	109.0	120.0	5.5	4.2	12.0	74.0	18.6	17.6	8.35	4	68	18
HE 120 A	19.9	114.0	120.0	8.0	5.0	12.0	74.0	25.3	14.8	5.69	5	68	20
HE 120 B	26.7	120.0	120.0	11.0	6.5	12.0	74.0	34.0	11.4	4.07	5	68	24
HE 120 C	39.2	130.0	123.0	16.0	9.5	12.0	74.0	49.9	7.79	2.80	7	68	28
HE 120 M	52.1	140.0	126.0	21.0	12.5	12.0	74.0	66.4	5.92	2.13	8	68	34
HE 140 AA	18.1	128.0	140.0	6.0	4.3	12.0	92.0	23.0	21.4	9.31	4	78	18
HE 140 A	24.7	133.0	140.0	8.5	5.5	12.0	92.0	31.4	16.7	6.50	5	78	22
HE 140 B	33.7	140.0	140.0	12.0	7.0	12.0	92.0	43.0	13.1	4.54	6	78	24
HE 140 C	48.2	150.0	143.0	17.0	10.0	12.0	92.0	61.5	9.20	3.21	7	78	30
HE 140 M	63.2	160.0	146.0	22.0	13.0	12.0	92.0	80.6	7.08	2.48	9	78	34
HE 160 AA	23.8	148.0	160.0	7.0	4.5	15.0	104.0	30.4	23.1	8.96	4	88	22
HE 160 A	30.4	152.0	160.0	9.0	6.0	15.0	104.0	38.8	17.3	6.89	5	88	24
HE 160 B	42.6	160.0	160.0	13.0	8.0	15.0	104.0	54.3	13.0	4.69	6	86	28
HE 160 C	59.2	170.0	163.0	18.0	11.0	15.0	104.0	75.4	9.45	3.39	8	86	34
HE 160 M	76.2	180.0	166.0	23.0	14.0	15.0	104.0	97.1	7.43	2.65	9	86	38
HE 180 AA	28.7	167.0	180.0	7.5	5.0	15.0	122.0	36.5	24.4	9.67	5	98	24
HE 180 A	35.5	171.0	180.0	9.5	6.0	15.0	122.0	45.3	20.3	7.58	5	98	26
HE 180 B	51.2	180.0	180.0	14.0	8.5	15.0	122.0	65.3	14.4	5.05	6	96	30
HE 180 C	69.8	190.0	183.0	19.0	11.5	15.0	122.0	89.0	10.6	3.72	8	96	34
HE 180 M	88.9	200.0	186.0	24.0	14.5	15.0	122.0	113	8.41	2.95	9	96	40
HE 200 AA	34.6	186.0	200.0	8.0	5.5	18.0	134.0	44.1	24.4	9.91	5	108	26

Dimensions	EN10365
Specification	EN10025
Size Range	HE 100 AA to HE 1000 x 584

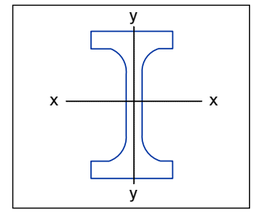


	Properties												
Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
Per metre									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
0.553	236	92.1	3.89	2.43	52.0	18	58.4	28.4	4.271	12.5	0.00168	2.33
0.561	349	134	4.06	2.51	72.8	27	83.0	41.1	4.363	9.95	0.00258	5.28
0.567	450	167	4.16	2.53	89.9	33	104	51.4	4.401	8.49	0.00338	9.33
0.593	759	274	4.39	2.64	138	53	166	82.1	4.502	6.23	0.00616	29.1
0.619	1,140	399	4.63	2.74	190	75	236	116	4.597	5.02	0.00993	67.2
0.669	413	159	4.72	2.93	75.8	27	84.1	40.6	4.690	15.7	0.00424	2.59
0.677	606	231	4.89	3.02	106	39	120	58.8	4.789	12.3	0.00647	6.04
0.686	864	318	5.04	3.06	144	53	165	81.0	4.840	9.62	0.00941	13.9
0.712	1,390	498	5.27	3.16	214	81	253	124	4.930	7.13	0.0161	40.7
0.738	2,020	703	5.51	3.25	288	112	351	172	5.016	5.75	0.0248	90.5
0.787	720	275	5.59	3.45	112	39	124	59.9	5.114	17.8	0.0102	3.43
0.794	1,030	389	5.73	3.52	155	56	174	84.8	5.175	13.9	0.0151	8.10
0.805	1,510	550	5.93	3.58	216	79	245	120	5.241	10.6	0.0225	20.2
0.831	2,330	830	6.16	3.68	311	116	364	178	5.323	7.92	0.0366	55.4
0.857	3,290	1,140	6.39	3.77	411	157	494	240	5.401	6.42	0.0543	119
0.901	1,280	479	6.50	3.97	173	60	190	91.4	5.504	17.3	0.0238	6.43
0.906	1,670	616	6.57	3.98	220	77	245	118	5.519	14.5	0.0314	12.1
0.918	2,490	889	6.78	4.05	312	111	354	170	5.589	10.9	0.0479	31.3
0.944	3,700	1,300	7.01	4.16	436	160	508	245	5.669	8.41	0.0750	78.7
0.970	5,100	1,760	7.25	4.26	566	212	675	326	5.747	6.89	0.108	161
1.02	1,970	730	7.34	4.47	236	81	258	124	5.846	18.9	0.0464	8.31
1.02	2,510	925	7.45	4.52	294	103	325	156	5.882	15.9	0.0602	14.9
1.04	3,830	1,360	7.66	4.57	426	151	481	231	5.941	11.7	0.0938	42.2
1.06	5,540	1,940	7.89	4.68	584	212	675	325	6.016	9.05	0.142	101
1.09	7,480	2,580	8.13	4.77	748	277	883	425	6.088	7.46	0.199	201
1.13	2,940	1,070	8.17	4.92	317	107	347	163	6.152	18.9	0.0845	12.5

2.2
Wide Flange Beams (HE)

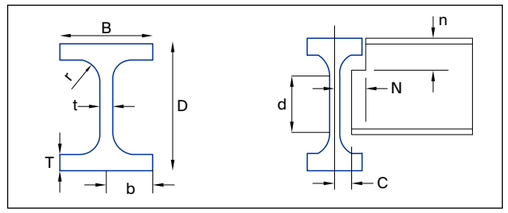


Dimensions	EN10365
Specification	EN10025
Size Range	HE 100 AA to HE 1000 x 584



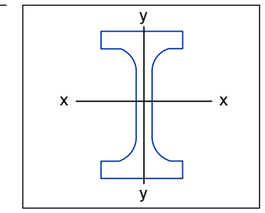
Section designation	Dimensions													Properties											
	Mass per Metre	Depth of Section	Width of Section	Thickness		Root	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing			Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
				Flange	Web				Flange	Web	End clearance	Notch		Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
	kg/m	D mm	B mm	T mm	t mm	r mm	d mm	A cm²	b/T	d/t	C mm	N mm	n mm	m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³	u	x	H dm⁶	J cm⁴
HE 200 A	42.3	190.0	200.0	10.0	6.5	18.0	134.0	53.8	20.6	7.88	5	108	28	1.14	3,690	1,340	8.28	4.98	389	134	430	204	6.194	16.3	0.108	21.0
HE 200 B	61.3	200.0	200.0	15.0	9.0	18.0	134.0	78.1	14.9	5.17	7	106	34	1.15	5,700	2,000	8.54	5.07	570	200	642	306	6.267	12.0	0.171	59.7
HE 200 C	81.9	210.0	203.0	20.0	12.0	18.0	134.0	104	11.2	3.88	8	106	38	1.18	8,030	2,790	8.77	5.17	765	275	881	421	6.339	9.45	0.252	135
HE 200 M	103	220.0	206.0	25.0	15.0	18.0	134.0	131	8.93	3.10	10	106	44	1.20	10,600	3,650	9.00	5.27	967	354	1,140	543	6.406	7.87	0.346	258
HE 220 AA	40.4	205.0	220.0	8.5	6.0	18.0	152.0	51.5	25.3	10.47	5	118	28	1.25	4,170	1,510	9.00	5.42	407	137	446	209	6.448	20.2	0.146	15.5
HE 220 A	50.5	210.0	220.0	11.0	7.0	18.0	152.0	64.3	21.7	8.05	6	118	30	1.26	5,410	1,960	9.17	5.51	515	178	568	271	6.517	16.9	0.193	28.6
HE 220 B	71.5	220.0	220.0	16.0	9.5	18.0	152.0	91.0	16.0	5.45	7	116	34	1.27	8,090	2,840	9.43	5.59	736	258	827	394	6.587	12.5	0.295	77.0
HE 220 C	94.1	230.0	223.0	21.0	12.5	18.0	152.0	120	12.2	4.15	8	116	40	1.30	11,200	3,890	9.65	5.69	972	349	1,110	532	6.652	9.98	0.424	168
HE 220 M	117	240.0	226.0	26.0	15.5	18.0	152.0	149	9.81	3.36	10	116	44	1.32	14,600	5,010	9.89	5.79	1,220	444	1,420	679	6.715	8.35	0.573	313
HE 240 AA	47.4	224.0	240.0	9.0	6.5	21.0	164.0	60.4	25.2	10.64	5	128	30	1.36	5,840	2,080	9.83	5.87	521	173	571	264	6.726	20.1	0.240	22.1
HE 240 A	60.3	230.0	240.0	12.0	7.5	21.0	164.0	76.8	21.9	7.94	6	128	34	1.37	7,760	2,770	10.1	6.00	675	231	745	352	6.815	16.6	0.328	42.1
HE 240 B	83.2	240.0	240.0	17.0	10.0	21.0	164.0	106	16.4	5.53	7	126	38	1.38	11,300	3,920	10.3	6.08	938	327	1,050	498	6.880	12.7	0.487	104
HE 240 C	119	255.0	244.0	24.5	14.0	21.0	164.0	152	11.7	3.84	9	126	46	1.42	17,300	5,940	10.7	6.25	1,360	487	1,560	744	6.984	9.45	0.788	289
HE 240 M	157	270.0	248.0	32.0	18.0	21.0	164.0	200	9.11	2.94	11	126	54	1.46	24,300	8,150	11.0	6.39	1,800	658	2,120	1,010	7.077	7.59	1.15	626
HE 260 AA	54.1	244.0	260.0	9.5	6.5	24.0	177.0	69.0	27.2	10.82	5	138	34	1.47	7,980	2,790	10.8	6.36	654	214	714	328	7.026	20.1	0.383	30.1
HE 260 A	68.2	250.0	260.0	12.5	7.5	24.0	177.0	86.8	23.6	8.18	6	138	38	1.48	10,400	3,670	11.0	6.50	836	282	920	430	7.112	17.0	0.516	54.2
HE 260 B	93.0	260.0	260.0	17.5	10.0	24.0	177.0	118	17.7	5.77	7	136	42	1.50	14,900	5,140	11.2	6.58	1,150	395	1,280	602	7.175	13.2	0.754	127
HE 260 C	132	275.0	264.0	25.0	14.0	24.0	177.0	168	12.6	4.04	9	136	50	1.54	22,600	7,680	11.6	6.75	1,640	582	1,880	888	7.272	9.95	1.20	340
HE 260 M	172	290.0	268.0	32.5	18.0	24.0	177.0	220	9.83	3.11	11	136	58	1.57	31,300	10,400	11.9	6.90	2,160	780	2,520	1,190	7.362	8.03	1.73	720
HE 280 AA	61.2	264.0	280.0	10.0	7.0	24.0	196.0	78.0	28.0	11.25	6	148	34	1.59	10,600	3,660	11.6	6.85	800	262	873	399	7.299	21.3	0.590	35.5
HE 280 A	76.4	270.0	280.0	13.0	8.0	24.0	196.0	97.3	24.5	8.62	6	146	38	1.60	13,700	4,760	11.9	7.00	1,010	340	1,110	518	7.379	18.0	0.785	63.5
HE 280 B	103	280.0	280.0	18.0	10.5	24.0	196.0	131	18.7	6.15	7	146	42	1.62	19,300	6,600	12.1	7.09	1,380	471	1,530	718	7.442	14.0	1.13	146
HE 280 C	145	295.0	284.0	25.5	14.5	24.0	196.0	185	13.5	4.34	9	146	50	1.66	28,800	9,750	12.5	7.26	1,950	687	2,220	1,050	7.540	10.6	1.77	385
HE 280 M	189	310.0	288.0	33.0	18.5	24.0	196.0	240	10.6	3.36	11	146	58	1.69	39,600	13,200	12.8	7.40	2,550	914	2,970	1,400	7.628	8.54	2.52	807
HE 300 AA	69.8	283.0	300.0	10.5	7.5	27.0	208.0	88.9	27.7	11.36	6	158	38	1.70	13,800	4,730	12.5	7.30	976	316	1,060	482	7.541	21.0	0.877	47.8
HE 300 A	88.3	290.0	300.0	14.0	8.5	27.0	208.0	112	24.5	8.48	6	156	42	1.72	18,300	6,310	12.7	7.49	1,260	421	1,380	641	7.646	17.7	1.20	87.8

2.2
Wide Flange Beams (HE)



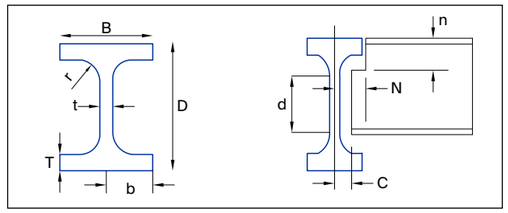
Section designation	Dimensions												
	Mass per Metre	Depth of Section	Width of Section	Thickness		Root	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
				Flange	Web				Flange	Web	End clearance	Notch	
	kg/m	D mm	B mm	T mm	t mm	r mm	d mm	A cm²	b/T	d/t	C mm	N mm	n mm
HE 300 B	117	300.0	300.0	19.0	11.0	27.0	208.0	149	18.9	6.18	8	156	46
HE 300 C	177	320.0	305.0	29.0	16.0	27.0	208.0	225	13.0	4.05	10	156	56
HE 300 M	238	340.0	310.0	39.0	21.0	27.0	208.0	303	9.90	3.01	13	156	66
HE 320 AA	74.2	301.0	300.0	11.0	8.0	27.0	225.0	94.6	28.1	10.82	6	156	38
HE 320 A	97.6	310.0	300.0	15.5	9.0	27.0	225.0	124	25.0	7.65	7	156	44
HE 320 B	127	320.0	300.0	20.5	11.5	27.0	225.0	161	19.6	5.72	8	156	48
HE 320 C	186	340.0	305.0	30.5	16.0	27.0	225.0	237	14.1	3.85	10	156	58
HE 320 M	245	359.0	309.0	40.0	21.0	27.0	225.0	312	10.7	2.93	13	154	68
HE 340 AA	78.9	320.0	300.0	11.5	8.5	27.0	243.0	100	28.6	10.33	6	156	40
HE 340 A	105	330.0	300.0	16.5	9.5	27.0	243.0	134	25.6	7.17	7	156	44
HE 340 B	134	340.0	300.0	21.5	12.0	27.0	243.0	171	20.3	5.44	8	154	50
HE 340 M	248	377.0	309.0	40.0	21.0	27.0	243.0	316	11.6	2.93	13	154	68
HE 360 AA	83.7	339.0	300.0	12.0	9.0	27.0	261.0	107	29.0	9.88	7	156	40
HE 360 A	112	350.0	300.0	17.5	10.0	27.0	261.0	143	26.1	6.74	7	156	46
HE 360 B	142	360.0	300.0	22.5	12.5	27.0	261.0	181	20.9	5.19	8	154	50
HE 360 M	250	395.0	308.0	40.0	21.0	27.0	261.0	319	12.4	2.91	13	154	68
HE 400 AA	92.4	378.0	300.0	13.0	9.5	27.0	298.0	118	31.4	9.10	7	156	40
HE 400 A	125	390.0	300.0	19.0	11.0	27.0	298.0	159	27.1	6.18	8	156	46
HE 400 B	155	400.0	300.0	24.0	13.5	27.0	298.0	198	22.1	4.84	9	154	52
HE 400 M	256	432.0	307.0	40.0	21.0	27.0	298.0	326	14.2	2.90	13	154	68
HE 450 AA	99.7	425.0	300.0	13.5	10.0	27.0	344.0	127	34.4	8.74	7	156	42
HE 450 A	140	440.0	300.0	21.0	11.5	27.0	344.0	178	29.9	5.58	8	156	48
HE 450 B	171	450.0	300.0	26.0	14.0	27.0	344.0	218	24.6	4.46	9	154	54
HE 450 M	263	478.0	307.0	40.0	21.0	27.0	344.0	335	16.4	2.90	13	154	68
HE 500 AA	107	472.0	300.0	14.0	10.5	27.0	390.0	137	37.1	8.41	7	156	42
HE 500 A	155	490.0	300.0	23.0	12.0	27.0	390.0	198	32.5	5.09	8	154	50

Dimensions	EN10365
Specification	EN10025
Size Range	HE 100 AA to HE 1000 x 584

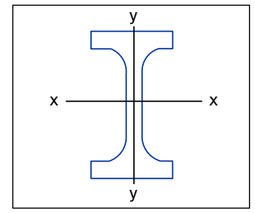


	Properties											
Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
1.73	25,200	8,560	13.0	7.58	1,680	571	1,870	870	7.706	14.1	1.69	189
1.78	41,000	13,700	13.5	7.81	2,560	901	2,930	1,370	7.835	10.0	2.90	604
1.83	59,200	19,400	14.0	8.00	3,480	1,250	4,080	1,910	7.947	7.88	4.39	1,410
1.74	16,400	4,960	13.2	7.24	1,090	331	1,200	506	7.617	21.8	1.04	53.6
1.76	22,900	6,980	13.6	7.49	1,480	466	1,630	710	7.761	17.5	1.51	112
1.77	30,800	9,240	13.8	7.57	1,930	616	2,150	939	7.813	14.2	2.07	230
1.82	48,700	14,400	14.3	7.81	2,860	947	3,270	1,440	7.945	10.3	3.45	688
1.87	68,100	19,700	14.8	7.95	3,800	1,280	4,440	1,950	8.028	8.20	5.00	1,510
1.78	19,600	5,180	14.0	7.18	1,220	346	1,340	529	7.675	22.6	1.23	60.0
1.79	27,700	7,440	14.4	7.46	1,680	496	1,850	756	7.837	17.9	1.82	131
1.81	36,700	9,690	14.7	7.53	2,160	646	2,410	986	7.882	14.5	2.45	263
1.90	76,400	19,700	15.6	7.90	4,050	1,280	4,720	1,950	8.085	8.70	5.58	1,510
1.81	23,000	5,410	14.7	7.12	1,360	361	1,500	553	7.710	23.3	1.44	67.1
1.83	33,100	7,890	15.2	7.43	1,890	526	2,090	802	7.893	18.1	2.18	153
1.85	43,200	10,100	15.5	7.49	2,400	676	2,680	1,030	7.935	14.8	2.88	298
1.93	84,900	19,500	16.3	7.83	4,300	1,270	4,990	1,940	8.115	9.21	6.14	1,510
1.89	31,200	5,860	16.3	7.06	1,650	391	1,820	600	7.778	24.8	1.95	81.3
1.91	45,100	8,560	16.8	7.34	2,310	571	2,560	873	7.951	19.0	2.94	193
1.93	57,700	10,800	17.1	7.40	2,880	721	3,230	1,100	7.987	15.7	3.82	361
2.00	104,000	19,300	17.9	7.70	4,820	1,260	5,570	1,930	8.155	10.3	7.41	1,520
1.98	41,900	6,090	18.2	6.92	1,970	406	2,180	624	7.780	27.4	2.57	91.4
2.01	63,700	9,460	18.9	7.29	2,900	631	3,220	966	8.016	20.0	4.15	250
2.03	79,900	11,700	19.1	7.33	3,550	781	3,980	1,200	8.039	16.7	5.26	448
2.10	132,000	19,300	19.8	7.59	5,500	1,260	6,330	1,940	8.176	11.6	9.25	1,530
2.08	54,600	6,310	20.0	6.79	2,320	421	2,580	649	7.756	29.9	3.30	103
2.11	87,000	10,400	21.0	7.24	3,550	691	3,950	1,060	8.050	20.8	5.64	318

2.2
Wide Flange Beams (HE)

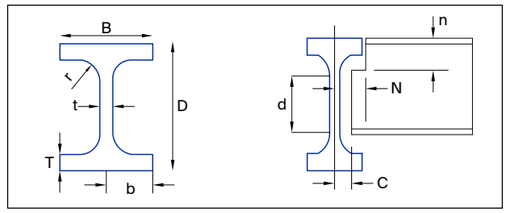


Dimensions	EN10365
Specification	EN10025
Size Range	HE 100 AA to HE 1000 x 584

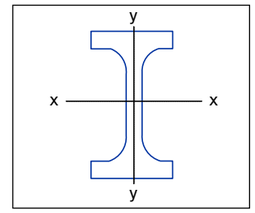


Section designation	Dimensions													Properties											
	Mass per Metre	Depth of Section	Width of Section	Thickness		Root	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing			Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
				Flange	Web				Flange	Web	End clearance	Notch														
		D	B	T	t	r	d	A	b/T	d/t	C	N	n	Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	u	x	H	J
	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm	m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
HE 500 B	187	500.0	300.0	28.0	14.5	27.0	390.0	239	26.9	4.13	9	154	56	2.12	107,000	12,600	21.2	7.27	4,290	842	4,820	1,290	8.065	17.6	7.02	548
HE 500 M	270	524.0	306.0	40.0	21.0	27.0	390.0	344	18.6	2.89	13	154	68	2.18	162,000	19,200	21.7	7.46	6,180	1,250	7,090	1,930	8.157	12.9	11.2	1,540
HE 550 AA	120	522.0	300.0	15.0	11.5	27.0	438.0	153	38.1	7.82	8	156	42	2.17	72,900	6,770	21.8	6.65	2,790	451	3,130	699	7.708	31.5	4.34	127
HE 550 A	166	540.0	300.0	24.0	12.5	27.0	438.0	212	35.0	4.86	8	154	52	2.21	112,000	10,800	23.0	7.15	4,150	721	4,620	1,110	8.031	22.4	7.19	360
HE 550 B	199	550.0	300.0	29.0	15.0	27.0	438.0	254	29.2	3.98	10	154	56	2.22	137,000	13,100	23.2	7.17	4,970	872	5,590	1,340	8.045	19.0	8.86	610
HE 550 M	278	572.0	306.0	40.0	21.0	27.0	438.0	354	20.9	2.89	13	154	68	2.28	198,000	19,200	23.6	7.35	6,920	1,250	7,930	1,940	8.135	14.3	13.5	1,560
HE 600 AA	129	571.0	300.0	15.5	12.0	27.0	486.0	164	40.5	7.55	8	154	44	2.27	91,900	6,990	23.7	6.53	3,220	466	3,620	724	7.651	33.8	5.38	142
HE 600 A	178	590.0	300.0	25.0	13.0	27.0	486.0	226	37.4	4.66	9	154	52	2.31	141,000	11,300	25.0	7.05	4,790	751	5,350	1,160	8.001	23.8	8.98	407
HE 600 B	212	600.0	300.0	30.0	15.5	27.0	486.0	270	31.4	3.84	10	154	58	2.32	171,000	13,500	25.2	7.08	5,700	902	6,420	1,390	8.015	20.3	11.0	677
HE 600 M	285	620.0	305.0	40.0	21.0	27.0	486.0	364	23.1	2.88	13	152	68	2.37	237,000	19,000	25.6	7.22	7,660	1,240	8,770	1,930	8.088	15.8	15.9	1,570
HE 600 x 337	337	632.0	310.0	46.0	25.5	27.0	486.0	429	19.1	2.51	15	154	74	2.41	283,000	22,900	25.7	7.31	8,960	1,480	10,400	2,310	8.123	13.9	19.6	2,440
HE 600 x 399	399	648.0	315.0	54.0	30.0	27.0	486.0	508	16.2	2.14	17	154	82	2.45	345,000	28,300	26.0	7.46	10,600	1,800	12,500	2,810	8.201	12.0	24.8	3,930
HE 650 AA	138	620.0	300.0	16.0	12.5	27.0	534.0	176	42.7	7.30	8	154	44	2.37	114,000	7,220	25.5	6.41	3,680	481	4,160	751	7.594	35.9	6.57	158
HE 650 A	190	640.0	300.0	26.0	13.5	27.0	534.0	242	39.6	4.47	9	154	54	2.41	175,000	11,700	26.9	6.97	5,470	782	6,140	1,200	7.966	25.2	11.0	458
HE 650 B	225	650.0	300.0	31.0	16.0	27.0	534.0	286	33.4	3.71	10	152	58	2.42	211,000	14,000	27.1	6.99	6,480	932	7,320	1,440	7.978	21.6	13.4	749
HE 650 M	293	668.0	305.0	40.0	21.0	27.0	534.0	374	25.4	2.88	13	152	68	2.47	282,000	19,000	27.5	7.13	8,430	1,240	9,660	1,940	8.048	17.2	18.6	1,580
HE 650 x 343	343	680.0	309.0	46.0	25.0	27.0	534.0	438	21.4	2.50	15	152	74	2.50	334,000	22,700	27.6	7.21	9,820	1,470	11,400	2,300	8.084	15.2	22.7	2,440
HE 650 x 407	407	696.0	314.0	54.0	29.5	27.0	534.0	519	18.1	2.13	17	154	82	2.54	405,000	28,000	28.0	7.35	11,600	1,780	13,600	2,800	8.160	13.2	28.7	3,930
HE 700 AA	150	670.0	300.0	17.0	13.0	27.0	582.0	191	44.8	6.85	9	154	44	2.47	143,000	7,670	27.3	6.34	4,260	512	4,840	800	7.562	37.3	8.16	186
HE 700 A	204	690.0	300.0	27.0	14.5	27.0	582.0	260	40.1	4.29	9	154	54	2.50	215,000	12,200	28.8	6.84	6,240	812	7,030	1,260	7.897	26.5	13.4	522
HE 700 B	241	700.0	300.0	32.0	17.0	27.0	582.0	306	34.2	3.58	11	152	60	2.52	257,000	14,400	29.0	6.87	7,340	963	8,330	1,500	7.912	22.8	16.1	839
HE 700 M	301	716.0	304.0	40.0	21.0	27.0	582.0	383	27.7	2.86	13	152	68	2.56	329,000	18,800	29.3	7.01	9,200	1,240	10,500	1,930	7.991	18.7	21.4	1,600
HE 700 x 352	352	728.0	308.0	46.0	25.0	27.0	582.0	449	23.3	2.49	15	152	74	2.59	390,000	22,500	29.5	7.08	10,700	1,460	12,400	2,290	8.027	16.5	26.0	2,450
HE 700 x 418	418	744.0	313.0	54.0	29.5	27.0	582.0	532	19.7	2.13	17	152	82	2.63	472,000	27,800	29.8	7.22	12,700	1,770	14,800	2,800	8.100	14.3	32.8	3,960
HE 800 AA	172	770.0	300.0	18.0	14.0	30.0	674.0	218	48.1	6.28	9	154	48	2.66	209,000	8,130	30.9	6.10	5,430	542	6,220	857	7.424	40.2	11.4	243

2.2
Wide Flange Beams (HE)

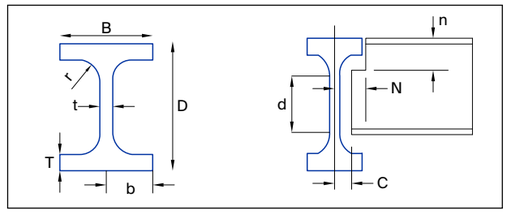


Dimensions	EN10365
Specification	EN10025
Size Range	HE 100 AA to HE 1000 x 584



Section designation	Dimensions													Properties												
	Mass per Metre	Depth of Section	Width of Section	Thickness		Root	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing			Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
				Flange	Web				Flange	Web	End clearance	Notch														
		D	B	T	t	r	d	A	b/T	d/t	C	N	n	Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	u	x	H	J
	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm	m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
HE 800 A	224	790.0	300.0	28.0	15.0	30.0	674.0	286	44.9	4.02	10	154	58	2.70	303,000	12,600	32.6	6.65	7,680	843	8,700	1,310	7.799	29.5	18.3	609
HE 800 B	262	800.0	300.0	33.0	17.5	30.0	674.0	334	38.5	3.37	11	152	64	2.71	359,000	14,900	32.8	6.68	8,980	994	10,200	1,550	7.813	25.6	21.8	959
HE 800 M	317	814.0	303.0	40.0	21.0	30.0	674.0	404	32.1	2.78	13	152	70	2.75	443,000	18,600	33.1	6.79	10,900	1,230	12,500	1,930	7.876	21.6	27.8	1,660
HE 800 x 373	373	826.0	308.0	46.0	25.0	30.0	674.0	475	27.0	2.42	15	152	76	2.78	524,000	22,500	33.2	6.89	12,700	1,460	14,700	2,310	7.927	19.0	34.1	2,550
HE 800 x 444	444	842.0	313.0	54.0	30.0	30.0	674.0	566	22.5	2.06	17	152	84	2.82	634,000	27,800	33.5	7.01	15,100	1,780	17,600	2,830	7.987	16.4	42.8	4,160
HE 900 AA	198	870.0	300.0	20.0	15.0	30.0	770.0	252	51.3	5.63	10	154	50	2.86	301,000	9,040	34.6	5.99	6,920	603	8,000	958	7.357	42.5	16.3	322
HE 900 A	252	890.0	300.0	30.0	16.0	30.0	770.0	320	48.1	3.73	10	152	60	2.90	422,000	13,600	36.3	6.50	9,480	903	10,800	1,410	7.711	31.8	25.0	749
HE 900 B	291	900.0	300.0	35.0	18.5	30.0	770.0	371	41.6	3.16	11	152	66	2.91	494,000	15,800	36.5	6.53	11,000	1,050	12,600	1,660	7.723	27.7	29.5	1,150
HE 900 M	333	910.0	302.0	40.0	21.0	30.0	770.0	424	36.7	2.76	13	152	70	2.93	570,000	18,400	36.7	6.60	12,500	1,220	14,400	1,930	7.766	24.6	34.8	1,680
HE 900 x 391	391	922.0	307.0	46.0	25.0	30.0	770.0	498	30.8	2.41	15	152	76	2.97	674,000	22,300	36.8	6.70	14,600	1,450	17,000	2,310	7.815	21.6	42.6	2,600
HE 900 x 466	466	938.0	312.0	54.0	30.0	30.0	770.0	594	25.7	2.06	17	152	84	3.01	815,000	27,600	37.1	6.81	17,400	1,770	20,400	2,830	7.877	18.7	53.4	4,230
HE 1000 AA	222	970.0	300.0	21.0	16.0	30.0	868.0	282	54.3	5.33	10	152	52	3.06	406,000	9,500	38.0	5.80	8,380	633	9,780	1,020	7.235	45.7	21.3	387
HE 1000 x 249	249	980.0	300.0	26.0	16.5	30.0	868.0	317	52.6	4.30	10	152	56	3.08	481,000	11,800	39.0	6.09	9,820	784	11,400	1,240	7.437	39.7	26.6	582
HE 1000 A	272	990.0	300.0	31.0	16.5	30.0	868.0	347	52.6	3.60	10	152	62	3.10	554,000	14,000	40.0	6.35	11,200	934	12,800	1,470	7.618	34.9	32.1	835
HE 1000 B	314	1,000.0	300.0	36.0	19.0	30.0	868.0	400	45.7	3.07	12	152	66	3.11	645,000	16,300	40.2	6.38	12,900	1,080	14,900	1,720	7.634	30.6	37.6	1,270
HE 1000 M	349	1,008.0	302.0	40.0	21.0	30.0	868.0	444	41.3	2.76	13	152	70	3.13	722,000	18,500	40.3	6.45	14,300	1,220	16,600	1,940	7.673	27.8	43.0	1,710
HE 1000 x 393	393	1,016.0	303.0	43.9	24.4	30.0	868.0	500	35.6	2.49	14	150	74	3.14	808,000	20,500	40.2	6.40	15,900	1,350	18,500	2,170	7.635	25.4	48.1	2,330
HE 1000 x 415	415	1,020.0	304.0	46.0	26.0	30.0	868.0	529	33.4	2.37	15	150	76	3.15	853,000	21,700	40.2	6.41	16,700	1,430	19,600	2,300	7.635	24.3	51.1	2,700
HE 1000 x 438	438	1,026.0	305.0	49.0	26.9	30.0	868.0	556	32.3	2.23	15	150	80	3.17	909,000	23,400	40.4	6.48	17,700	1,530	20,800	2,460	7.684	23.0	55.3	3,180
HE 1000 x 494	494	1,036.0	309.0	54.0	31.0	30.0	868.0	629	28.0	2.02	18	150	84	3.19	1,028,000	26,800	40.4	6.53	19,800	1,740	23,400	2,820	7.702	20.9	64.0	4,400
HE 1000 x 584	584	1,056.0	314.0	64.0	36.0	30.0	868.0	744	24.1	1.70	20	150	94	3.24	1,246,000	33,400	40.9	6.70	23,600	2,130	28,000	3,480	7.805	18.0	81.2	7,150

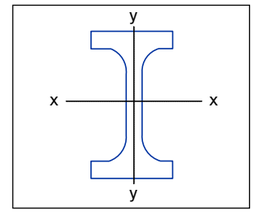
2.3 Universal Beams & Columns (UB & UC)



Imperial Units

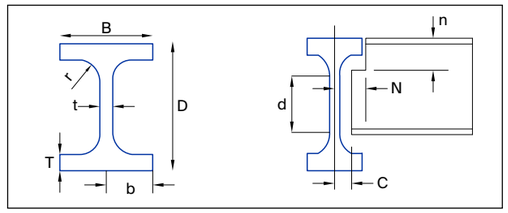
Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W4-4x4 (102x102)	13	19.4	105.7	103.1	8.8	7.1	6.4	75.5	24.7	5.88	10.6	6	58	15
	13.8	21.0	102.0	102.0	9.4	8.0	6.0	71.2	26.1	5.43	8.90	6	57	15
	16.3	24.0	107.0	100.0	12.0	7.9	6.0	71.0	30.9	4.17	8.99	6	56	18
W5-5x3 (127x76)	9	13.0	127.0	76.0	7.6	4.0	7.6	96.6	16.5	5.00	24.2	4	46	15
W5-5x5 (127x127)	16	23.8	127.3	127.0	9.1	6.1	7.6	93.8	30.4	6.95	15.4	5	70	17
	19	28.3	130.8	127.8	10.9	6.9	7.6	93.8	35.8	5.86	13.7	5	70	19
W6-6x3 (152x76)	9	14.0	150.0	75.0	7.0	5.0	8.0	120.0	17.8	5.36	24.0	5	45	15
	12	18.0	155.0	75.0	8.5	6.0	8.0	122.0	21.6	4.41	20.3	5	45	17
W6-6x3 1/2 (152x89)	11	16.0	152.4	88.7	7.7	4.5	7.6	121.8	20.3	5.76	27.1	4	52	15
W6-6x4 (152x102)	9	13.4	149.9	100.1	5.5	4.3	6.4	126.3	17.3	9.17	29.2	4	58	12
	12	17.9	153.2	101.6	7.1	5.8	6.4	126.3	22.9	7.14	21.6	5	58	13
	16	23.8	159.5	102.4	10.2	6.6	6.4	126.4	30.4	5.02	19.2	5	58	17
W6-6x6 (152X152)	15	22.3	152.1	152.1	6.6	5.8	6.4	126.2	28.5	11.5	21.6	5	83	13
	15.7	23.0	152.4	152.2	6.8	5.8	7.6	123.6	29.2	11.2	21.3	5	83	14
	20	30.0	157.6	152.9	9.4	6.5	7.6	123.6	38.3	8.13	19.0	5	83	17
	25	37.0	161.8	154.4	11.5	8.0	7.6	123.6	47.1	6.71	15.5	6	83	19
W7-7x 3 1/2 (178x89)	11	16.1	173.0	90.0	7.0	4.5	8.9	141.2	20.4	6.43	31.4	4	53	16
	12	18.1	175.0	90.0	8.0	5.0	8.9	141.2	23.0	5.63	28.2	5	53	17
	15	22.2	179.0	90.0	10.0	6.0	8.9	141.2	28.2	4.50	23.5	5	52	19
W7-7x4 (178x102)	13	19.0	177.8	101.2	7.9	4.8	7.6	146.8	24.3	6.41	30.6	4	58	16
W8-8x4 (203x102)	10	14.9	200.4	100.1	5.2	4.3	7.6	174.7	19.1	9.61	40.4	4	58	13
	13	19.4	202.9	101.6	6.5	5.8	7.6	174.7	24.8	7.84	29.9	5	58	14
	15	22.3	206.0	102.0	8.0	6.2	7.6	174.8	28.6	6.38	28.1	5	58	16
	16	23.1	203.2	101.8	9.3	5.4	7.6	169.4	29.4	5.47	31.4	5	58	17
W8-8x5 1/4 (203x133)	15	22.3	202.0	133.0	7.0	5.0	8.9	170.2	28.7	9.50	34.0	5	74	16
	17	25.1	203.2	133.2	7.8	5.7	7.6	172.4	32.0	8.54	30.2	5	74	15
	18	26.8	206.8	133.3	8.4	5.8	7.6	174.8	33.9	7.95	29.9	5	74	16
	20	30.0	206.8	133.9	9.6	6.4	7.6	172.4	38.2	6.97	26.9	5	74	17
	21	31.3	210.3	133.9	10.1	6.4	7.6	174.9	39.6	6.63	27.5	5	74	18

Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)

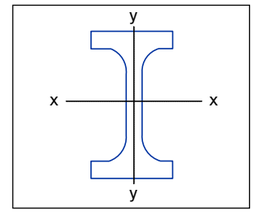


Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
0.599	472	160	4.38	2.55	89.4	31.1	103	47.8	0.836	10.9	0.004	6.29
0.586	456	167	4.18	2.53	89.4	32.7	104	50.4	0.827	9.61	0.004	7.78
0.588	587	200	4.36	2.55	110	40.1	129	61.5	0.844	8.13	0.005	13.5
0.537	473	55.8	5.35	1.84	74.6	14.7	84.2	22.6	0.895	16.3	0.002	2.85
0.737	892	312	5.42	3.21	140	49.2	158	75.0	0.842	13.0	0.011	8.00
0.746	1092	380	5.52	3.25	167	59.4	190	90.6	0.845	11.2	0.014	13.1
0.576	666	49.5	6.11	1.67	88.8	13.2	102	20.8	0.876	20.4	0.003	2.81
0.584	841	60.2	6.24	1.67	109	16.0	126	25.4	0.875	17.6	0.003	4.81
0.638	834	89.8	6.41	2.10	109	20.2	123	31.2	0.890	19.6	0.005	3.56
0.681	683	91.4	6.29	2.30	91.1	18.3	102	28.1	0.873	26.2	0.005	1.69
0.690	918	125	6.33	2.33	120	24.5	136	38.0	0.869	20.4	0.007	3.76
0.704	1330	183	6.61	2.45	167	35.7	190	55.2	0.882	15.4	0.010	9.12
0.890	1210	387	6.51	3.68	159	50.9	177	77.7	0.837	21.4	0.021	4.21
0.889	1250	400	6.54	3.70	164	52.6	182	80.2	0.840	20.7	0.021	4.63
0.901	1748	560	6.76	3.83	222	73.3	248	112	0.849	16.0	0.031	10.5
0.912	2210	706	6.85	3.87	273	91.5	309	140	0.848	13.3	0.040	19.2
0.682	1060	85.4	7.20	2.04	123	19.0	138	29.4	0.884	23.9	0.006	3.15
0.685	1213	97.6	7.26	2.06	139	21.7	157	33.7	0.885	21.4	0.007	4.48
0.691	1529	122	7.36	2.08	171	27.1	195	42.3	0.886	17.8	0.009	8.16
0.738	1356	137	7.48	2.37	153	27.0	171	41.6	0.888	22.6	0.010	4.41
0.779	1284	87.3	8.19	2.14	128	17.4	145	27.2	0.867	36.3	0.008	1.78
0.787	1647	114	8.16	2.14	162	22.4	187	35.3	0.861	29.0	0.011	3.63
0.794	1999	142	8.36	2.23	194	27.8	222	43.7	0.869	25.2	0.014	5.68
0.790	2105	164	8.46	2.36	207	32.2	234	49.8	0.888	22.5	0.015	7.02
0.911	2105	275	8.56	3.09	208	41.3	232	63.4	0.879	27.9	0.026	4.50
0.915	2340	308	8.56	3.10	230	46.2	258	70.9	0.877	25.6	0.029	5.96
0.922	2578	331	8.72	3.12	249	49.7	279	76.3	0.880	24.5	0.033	7.13
0.923	2896	385	8.71	3.17	280	57.5	314	88.2	0.881	21.5	0.037	10.3
0.930	3119	405	8.87	3.20	297	60.4	333	92.7	0.885	21.0	0.041	11.6

2.3
Universal Beams & Columns (UB & UC)



Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)

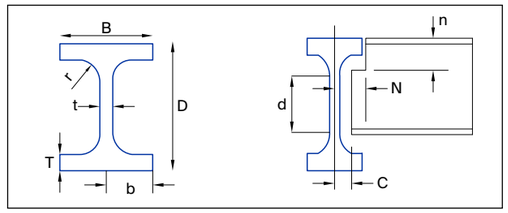


Imperial Units

Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W8-8x6 1/2 (203x165)	24	35.7	201.4	165.0	10.1	6.2	10.2	160.8	45.5	8.17	25.9	5	89	20
	28	41.7	204.7	166.0	11.8	7.2	10.2	160.7	53.2	7.03	22.2	6	89	22
W8-8x8 (203x203)	31	46.1	203.2	203.6	11.0	7.2	10.2	160.8	58.7	9.25	22.3	6	108	21
	35	52.0	206.2	204.3	12.5	7.9	10.2	160.8	66.3	8.17	20.4	6	108	23
	40	60.0	209.6	205.8	14.2	9.4	10.2	160.8	76.4	7.25	17.1	7	108	24
	48	71.0	215.8	206.4	17.3	10.0	10.2	160.8	90.4	5.97	16.1	7	108	28
	58	86.1	222.2	209.1	20.5	12.7	10.2	160.8	110	5.10	12.7	8	108	31
	67	99.7	228.6	210.3	23.8	14.5	10.2	160.7	127	4.43	11.1	9	108	34
W10-10x4 (254x102)	12	17.9	250.7	100.6	5.3	4.8	7.6	224.8	22.8	9.44	46.5	4	58	13
	15	22.0	254.0	101.6	6.8	5.7	7.6	225.2	28.0	7.47	39.5	5	58	14
	17	25.2	257.2	101.9	8.4	6.0	7.6	225.2	32.0	6.07	37.5	5	58	16
	19	28.3	260.4	102.2	10.0	6.3	7.6	225.2	36.1	5.11	35.7	5	58	18
W10-10x5 3/4 (254x146)	21	31.1	251.4	146.1	8.6	6.0	7.6	219.0	39.7	8.49	36.5	5	80	16
	22	32.7	258.3	146.0	9.1	6.1	7.6	224.8	41.8	7.99	36.8	5	80	17
	25	37.0	256.0	146.4	10.9	6.3	7.6	219.0	47.2	6.72	34.8	5	80	19
	26	38.7	262.4	146.6	11.1	6.6	7.6	225.0	48.9	6.60	34.1	5	80	19
	29	43.0	259.6	147.3	12.7	7.2	7.6	219.0	54.8	5.80	30.4	6	80	20
	30	44.6	265.9	147.6	12.9	7.6	7.6	224.9	56.9	5.72	29.5	6	80	21
W10-10x8 (254x203)	33	49.1	247.1	202.2	11.0	7.4	12.7	199.7	62.5	9.19	27.1	6	107	24
	39	58.0	252.0	202.8	13.4	8.0	12.7	199.8	73.8	7.57	25.0	6	107	26
	42	62.5	249.4	207.0	12.2	12.2	12.7	199.6	79.3	8.49	16.4	8	107	25
	45	67.0	256.5	203.7	15.7	8.9	12.7	199.7	85.4	6.49	22.5	6	107	28
	57	84.8	258.1	211.3	16.5	16.5	12.7	199.7	108	6.40	12.1	10	107	29
W10-10x10 (254x254)	49	73.1	254.1	254.6	14.2	8.6	12.7	200.3	93.1	8.96	23.3	6	133	27
	54	80.4	256.3	254.8	15.6	9.4	12.7	199.7	102	8.17	21.2	7	133	28
	60	88.9	260.3	256.3	17.3	10.3	12.7	200.3	113	7.41	19.4	7	133	30
	68	101.2	264.2	257.3	19.6	11.9	12.7	199.7	129	6.58	16.7	8	133	32
	72	107.1	266.7	258.8	20.5	12.8	12.7	200.3	136	6.31	15.6	8	133	33
	77	114.6	269.2	258.8	22.1	13.5	12.7	199.6	146	5.86	14.8	9	133	35
	88	131	275.3	260.7	25.2	15.4	12.7	199.6	167	5.18	13.0	10	133	38
	89	132	276.3	261.3	25.3	15.3	12.7	200.3	168	5.16	13.1	10	133	38
	100	148.8	281.9	262.6	28.5	17.3	12.7	199.6	190	4.62	11.6	11	133	41
	112	167.1	289.1	265.2	31.7	19.2	12.7	200.3	213	4.18	10.4	12	133	44

Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³	u	x	H	J
1.03	3430	757	8.68	4.08	341	91.7	378	140	0.875	19.3	0.069	14.3
1.04	4077	901	8.76	4.12	398	109	445	165	0.875	16.8	0.084	22.4
1.19	4568	1548	8.82	5.13	450	152	497	231	0.847	17.7	0.143	22.2
1.20	5259	1778	8.91	5.18	510	174	567	264	0.848	15.8	0.167	31.8
1.21	6125	2065	8.96	5.20	584	201	656	305	0.846	14.1	0.197	47.2
1.22	7618	2537	9.18	5.30	706	246	799	374	0.853	11.9	0.250	80.2
1.24	9449	3128	9.28	5.34	850	299	977	456	0.850	10.2	0.318	137
1.25	11310	3687	9.44	5.39	990	351	1150	536	0.852	9.0	0.387	211
0.881	2241	90.8	9.91	1.99	179	18.0	207	28.6	0.851	44.0	0.014	2.28
0.890	2841	119	10.1	2.06	224	23.5	259	37.3	0.856	36.4	0.018	4.15
0.897	3415	149	10.3	2.15	266	29.2	306	46.0	0.866	31.5	0.023	6.42
0.904	4005	179	10.5	2.22	308	34.9	353	54.8	0.874	27.5	0.028	9.57
1.06	4413	448	10.5	3.36	351	61.3	393	94.1	0.880	29.6	0.066	8.55
1.08	4917	475	10.8	3.37	381	65.0	426	100	0.882	28.9	0.074	9.96
1.07	5537	571	10.8	3.48	433	78.0	483	119	0.890	24.3	0.086	15.3
1.08	5974	584	11.1	3.45	455	79.6	510	122	0.888	24.5	0.092	16.5
1.08	6544	677	10.9	3.52	504	92.0	566	141	0.891	21.2	0.103	23.9
1.09	7048	692	11.1	3.49	530	93.8	597	144	0.888	21.3	0.111	25.6
1.27	7071	1517	10.6	4.93	572	150	634	229	0.873	21.5	0.211	24.0
1.28	8672	1865	10.8	5.03	688	184	765	280	0.878	18.3	0.265	40.2
1.25	8430	1807	10.3	4.77	676	175	768	271	0.851	17.9	0.254	44.5
1.29	10300	2214	11.0	5.09	803	217	898	331	0.880	15.9	0.321	62.3
1.31	11930	2606	10.5	4.91	925	247	1067	385	0.849	13.6	0.380	109
1.49	11410	3908	11.1	6.48	898	307	992	465	0.849	17.3	0.562	57.6
1.49	12590	4304	11.1	6.49	983	338	1091	512	0.849	15.8	0.623	75.6
1.50	14270	4858	11.2	6.55	1096	379	1224	575	0.850	14.5	0.717	102
1.51	16390	5558	11.3	6.57	1241	432	1398	657	0.849	12.9	0.832	148
1.52	17510	5928	11.3	6.59	1313	458	1484	697	0.848	12.4	0.898	172
1.52	18950	6391	11.4	6.61	1408	494	1599	752	0.849	11.6	0.976	213
1.54	22210	7436	11.5	6.67	1613	570	1850	869	0.850	10.3	1.16	314
1.55	22530	7531	11.6	6.69	1631	576	1869	878	0.850	10.3	1.19	319
1.56	25900	8598	11.7	6.73	1838	655	2127	999	0.851	9.3	1.38	452
1.58	30000	9870	11.9	6.81	2075	744	2424	1137	0.851	8.5	1.63	626

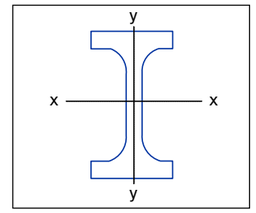
2.3
Universal Beams & Columns (UB & UC)



Imperial Units

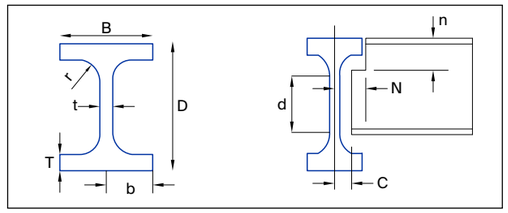
Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W12-12x4 (305x102)	14	20.8	302.5	100.8	5.7	5.1	7.6	275.8	26.8	8.83	54.3	5	58	13
	16	23.8	304.5	101.3	6.7	5.6	7.6	275.8	30.4	7.53	49.3	5	58	14
	16.5	24.8	305.1	101.6	7.0	5.8	7.6	275.9	31.6	7.26	47.6	5	58	15
	19	28.2	308.7	101.8	8.8	6.0	7.6	275.9	35.9	5.78	46.0	5	58	16
	22	32.8	312.7	102.4	10.8	6.6	7.6	275.9	41.8	4.74	41.8	5	58	18
W12-12x5 (305x127)	25	37.0	304.4	123.3	10.7	7.1	8.9	265.2	47.2	5.76	37.4	6	68	20
	28	41.9	307.2	124.3	12.1	8.0	8.9	265.2	53.4	5.14	33.2	6	68	21
	32	48.1	311.0	125.3	14.0	9.0	8.9	265.2	61.2	4.48	29.5	7	68	23
W12-12x6 1/2 (305x165)	26	38.7	310.4	164.8	9.7	5.8	7.6	275.9	49.3	8.54	47.2	5	89	17
	27	40.3	303.4	165.0	10.2	6.0	8.9	265.2	51.3	8.09	44.2	5	90	19
	30	44.6	313.4	165.6	11.1	6.6	7.6	276.0	56.5	7.46	41.8	5	90	19
	31	46.1	306.6	165.7	11.8	6.7	8.9	265.2	58.7	7.02	39.6	5	90	21
	35	52.1	317.5	166.6	13.2	7.6	7.6	275.9	66.7	6.31	36.2	6	89	21
	36	54.0	310.4	166.9	13.7	7.9	8.9	265.2	68.8	6.09	33.6	6	90	23
W12-12x8 (305x203)	40	59.5	303.3	203.3	13.0	7.5	15.2	246.9	75.6	7.82	33.0	6	108	28
	45	67.0	306.3	204.3	14.6	8.5	15.2	246.7	85.2	7.00	29.0	6	108	30
	50	74.4	309.6	205.2	16.2	9.4	15.2	246.8	94.5	6.33	26.3	7	108	31
W12-12x10 (305x254)	53	78.9	306.3	253.9	14.6	8.8	15.2	246.7	100	8.70	28.2	6	133	30
	58	86.3	309.6	254.3	16.2	9.1	15.2	246.8	110	7.85	27.0	7	133	31
W12-12x12 (305x305)	65	96.9	307.9	305.3	15.4	9.9	15.2	246.7	123	9.91	24.9	7	158	31
	72	107.1	311.1	305.8	17.0	10.9	15.2	246.7	136	8.98	22.6	7	157	32
	79	117.9	314.5	307.4	18.7	12.0	15.2	246.7	150	8.22	20.6	8	158	34
	87	129.5	318.3	308.0	20.6	13.1	15.2	246.8	165	7.49	18.9	9	157	36
	92	136.9	320.5	309.2	21.7	13.8	15.2	246.7	174	7.12	17.9	9	158	37
	96	142.8	322.8	308.9	22.9	14.0	15.2	246.7	182	6.76	17.7	9	157	38
	106	158.1	327.1	311.2	25.0	15.8	15.2	246.7	201	6.22	15.6	10	158	40
	120	178.6	333.2	312.9	28.1	18.0	15.2	246.7	228	5.57	13.7	11	157	43
	133	198.1	339.9	314.5	31.4	19.1	15.2	246.7	252	5.01	12.9	12	158	47
	136	202.4	340.6	315.0	31.8	20.1	15.2	246.7	258	4.96	12.3	12	157	47
	152	226.2	348.2	317.0	35.6	22.1	15.2	246.7	289	4.46	11.2	13	157	51
	161	240	352.5	318.4	37.7	23.0	15.2	246.7	306	4.22	10.7	14	158	53
	170	253	356.4	319.3	39.6	24.4	15.2	246.8	323	4.03	10.1	14	157	55

Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)



Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
0.985	3682	97.9	11.7	1.91	243	19.4	286	31.1	0.842	50.8	0.022	2.93
0.990	4274	117	11.9	1.96	281	23.1	329	37.0	0.846	45.0	0.026	4.27
0.992	4455	123	11.9	1.97	292	24.2	342	38.8	0.846	43.4	0.027	4.77
1.00	5366	155	12.2	2.08	348	30.5	403	48.5	0.859	37.4	0.035	7.40
1.01	6501	194	12.5	2.15	416	37.9	481	60.0	0.866	31.6	0.044	12.2
1.07	7166	335	12.3	2.67	471	54.4	539	85.3	0.872	29.7	0.072	14.8
1.08	8196	389	12.4	2.70	534	62.6	614	98.4	0.872	26.5	0.085	21.1
1.09	9575	461	12.5	2.74	616	73.6	711	116	0.873	23.3	0.102	31.8
1.26	8498	720	13.1	3.82	548	87.4	609	134	0.887	33.8	0.163	12.5
1.24	8503	764	12.9	3.86	560	92.7	623	142	0.889	31.0	0.164	14.7
1.26	9864	841	13.2	3.86	629	102	703	156	0.887	29.7	0.192	18.7
1.25	9899	896	13.0	3.90	646	108	720	166	0.891	27.1	0.195	22.2
1.27	11860	1019	13.3	3.91	747	122	838	188	0.889	25.3	0.236	30.8
1.26	11700	1063	13.0	3.93	754	127	846	196	0.889	23.6	0.234	34.8
1.38	12840	1823	13.0	4.91	847	179	938	274	0.890	22.9	0.384	39.1
1.39	14570	2078	13.1	4.94	952	203	1060	311	0.889	20.6	0.442	54.6
1.40	16350	2337	13.2	4.97	1056	228	1183	349	0.889	18.8	0.503	73.5
1.58	17700	3986	13.3	6.30	1156	314	1276	477	0.876	20.4	0.848	65.6
1.59	19740	4444	13.4	6.36	1275	349	1411	531	0.878	18.7	0.956	86.5
1.79	22250	7308	13.4	7.69	1445	479	1592	726	0.850	19.3	1.56	91.2
1.80	24830	8117	13.5	7.72	1596	531	1767	806	0.850	17.6	1.75	122
1.81	27670	9060	13.6	7.77	1760	589	1958	895	0.850	16.2	1.98	161
1.82	30810	10030	13.7	7.80	1936	651	2164	990	0.851	14.9	2.22	212
1.82	32820	10700	13.7	7.83	2048	692	2297	1053	0.851	14.2	2.39	249
1.83	34670	11240	13.8	7.86	2148	728	2413	1106	0.853	13.6	2.53	285
1.84	38750	12570	13.9	7.90	2369	808	2680	1230	0.851	12.5	2.87	378
1.86	44560	14350	14.0	7.94	2675	917	3053	1399	0.851	11.2	3.34	537
1.87	50900	16300	14.2	8.04	2995	1037	3440	1581	0.853	10.2	3.88	734
1.87	51790	16560	14.2	8.02	3041	1052	3501	1606	0.852	10.1	3.95	771
1.89	59610	18910	14.4	8.09	3424	1193	3975	1823	0.853	9.2	4.62	1074
1.91	64200	20320	14.5	8.15	3643	1276	4247	1951	0.854	8.7	5.03	1271
1.92	68490	21540	14.6	8.17	3844	1349	4502	2064	0.854	8.4	5.40	1480

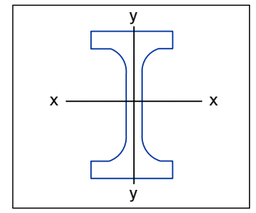
2.3
Universal Beams & Columns (UB & UC)



Imperial Units

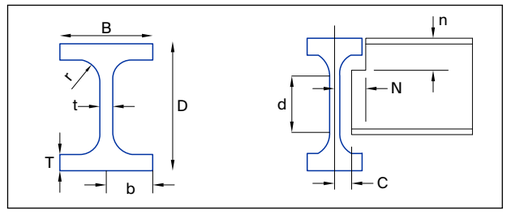
Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W12-12x12 (305x305)	190	282.9	365.3	322.2	44.1	26.8	15.2	246.7	360	3.65	9.21	15	158	59
	210	312.5	373.6	324.9	48.3	30.0	15.2	246.7	399	3.37	8.23	17	157	63
	230	342	382.3	327.5	52.6	32.6	15.2	246.7	437	3.11	7.57	18	157	68
	252	375	391.4	330.3	57.2	35.4	15.2	246.6	478	2.89	6.97	20	157	72
	278	413.7	402.6	333.8	62.7	38.9	15.2	246.7	529	2.66	6.35	21	157	78
	279	415	402.6	333.8	62.7	38.9	15.2	246.8	528	2.66	6.34	21	157	78
	305	454	414.5	336.2	68.7	41.3	15.2	246.7	578	2.45	5.97	23	157	84
	336	500	427.2	340.0	75.1	45.1	15.2	246.6	638	2.26	5.47	25	157	90
W14-14x5 (356x127)	22	33.1	349.0	125.4	8.5	6.0	10.2	311.6	42.1	7.38	51.9	5	70	19
	26	39.1	353.4	126.0	10.7	6.6	10.2	311.6	49.8	5.89	47.2	5	70	21
W14-14x6 3/4 (356x171)	30	45.0	351.4	171.1	9.7	7.0	10.2	311.6	57.3	8.82	44.5	6	92	20
	34	51.0	355.0	171.5	11.5	7.4	10.2	311.6	64.9	7.46	42.1	6	92	22
	38	57.0	358.0	172.2	13.0	8.1	10.2	311.6	72.6	6.62	38.5	6	92	23
	45	67.1	363.4	173.2	15.7	9.1	10.2	311.6	85.5	5.52	34.2	7	92	26
W14-14x8 (356x203)	43	64.0	347.0	203.1	13.4	7.8	15.2	289.8	81.2	7.58	37.4	6	108	29
	48	71.4	350.3	204.0	15.1	8.6	15.2	289.7	91.2	6.75	33.5	6	108	30
	53	78.9	353.6	204.7	16.7	9.4	15.2	289.8	100	6.13	30.8	7	108	32
W14-14x10 (356x254)	61	90.8	352.8	253.9	16.3	9.5	15.2	289.8	115	7.79	30.4	7	132	32
	68	101.2	356.6	254.8	18.3	10.5	15.2	289.6	129	6.97	27.5	7	132	33
	74	110.1	359.9	255.8	19.9	11.4	15.2	289.6	141	6.41	25.3	8	132	35
	82	122	363.5	257.3	21.7	12.9	15.2	289.7	155	5.92	22.5	8	132	37
W14-14x14 1/2 (356x368)	87	129	355.6	368.6	17.5	10.4	15.2	290.2	164	10.5	27.9	7	189	33
	90	133.9	356.1	368.8	18.0	11.2	15.2	289.6	171	10.2	25.9	8	189	33
	99	147.3	359.7	370.0	19.8	12.3	15.2	289.7	188	9.34	23.5	8	189	35
	103	152.9	362.0	370.5	20.7	12.3	15.2	290.2	195	8.95	23.6	8	189	36
	109	162.2	363.7	371.0	21.8	13.3	15.2	289.6	207	8.49	21.7	9	189	37
	119	177	368.2	372.6	23.8	14.4	15.2	290.2	226	7.83	20.2	9	189	39
	120	178.6	367.8	372.6	23.9	15.0	15.2	289.6	228	7.80	19.3	9	189	39
	132	196.4	372.4	374.0	26.2	16.4	15.2	289.7	250	7.15	17.7	10	189	41
	136	201.9	374.6	374.7	27.0	16.5	15.2	290.2	257	6.94	17.6	10	189	42
W14-14x16 (356x406)	118	175.6	360.0	396.0	20.0	20.0	15.0	290.0	224	9.90	14.5	12	198	35
	145	215.8	375.4	393.7	27.7	17.3	15.2	289.6	275	7.11	16.8	11	198	43
	158	235.1	381.0	394.8	30.2	18.4	15.2	290.2	299	6.54	15.8	11	198	45
	159	236.6	380.5	395.4	30.2	18.9	15.2	289.6	302	6.54	15.3	11	198	45

Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)



Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
1.94	78870	24640	14.8	8.27	4318	1529	5105	2342	0.855	7.7	6.35	2034
1.96	89270	27660	15.0	8.33	4779	1702	5703	2613	0.855	7.1	7.32	2693
1.98	100600	30880	15.2	8.41	5261	1886	6332	2898	0.856	6.6	8.39	3490
2.01	113200	34470	15.4	8.49	5783	2087	7020	3211	0.856	6.2	9.62	4495
2.04	129600	39040	15.7	8.59	6437	2339	7890	3604	0.857	5.7	11.3	5953
2.04	129500	39010	15.7	8.59	6435	2338	7888	3602	0.857	5.7	11.3	5946
2.07	147600	43690	16.0	8.69	7122	2599	8807	4006	0.860	5.4	13.1	7719
2.10	169000	49420	16.3	8.80	7913	2907	9882	4487	0.861	5.0	15.3	10110
1.17	8250	280	14.0	2.58	473	44.7	543	70.3	0.863	42.2	0.081	8.79
1.18	10170	358	14.3	2.68	576	56.8	659	89.1	0.871	35.2	0.105	15.1
1.36	12070	811	14.5	3.76	687	94.8	775	147	0.874	36.8	0.237	15.8
1.36	14140	968	14.8	3.86	796	113	896	174	0.881	32.1	0.286	23.8
1.37	16040	1108	14.9	3.91	896	129	1010	199	0.882	28.8	0.330	33.4
1.38	19460	1362	15.1	3.99	1071	157	1211	243	0.886	24.4	0.412	55.7
1.46	17760	1874	14.8	4.80	1024	185	1138	283	0.891	25.9	0.521	43.2
1.47	20170	2140	14.9	4.84	1151	210	1285	322	0.891	23.3	0.601	60.4
1.48	22480	2391	15.0	4.88	1271	234	1424	359	0.892	21.3	0.679	80.1
1.68	26540	4451	15.2	6.21	1505	351	1668	534	0.886	21.5	1.26	90.3
1.69	30060	5048	15.3	6.26	1686	396	1878	604	0.886	19.4	1.44	126
1.69	33120	5569	15.3	6.29	1840	435	2058	665	0.886	18.0	1.61	161
1.70	36680	6175	15.4	6.31	2018	480	2270	734	0.884	16.6	1.80	211
2.14	40250	14610	15.6	9.43	2264	793	2479	1199	0.844	19.9	4.18	153
2.14	41580	15080	15.6	9.40	2335	818	2565	1238	0.842	19.2	4.31	169
2.15	46240	16730	15.7	9.43	2571	904	2838	1370	0.842	17.7	4.83	223
2.16	48590	17550	15.8	9.49	2685	948	2965	1435	0.844	17.0	5.11	251
2.16	51540	18600	15.8	9.49	2834	1003	3142	1519	0.843	16.2	5.43	296
2.17	57120	20530	15.9	9.54	3103	1102	3455	1671	0.844	15.0	6.09	381
2.17	57290	20600	15.9	9.51	3115	1106	3475	1678	0.842	14.9	6.09	390
2.18	63720	22820	16.0	9.55	3422	1221	3838	1853	0.843	13.7	6.84	511
2.19	66260	23690	16.1	9.60	3538	1264	3972	1920	0.844	13.4	7.16	558
2.24	51770	20730	15.2	9.61	2876	1047	3235	1603	0.810	16.1	5.99	322
2.26	71240	28180	16.1	10.1	3796	1432	4264	2172	0.832	13.0	8.52	632
2.28	79090	30990	16.3	10.2	4151	1570	4687	2383	0.834	12.1	9.54	812
2.28	79160	31170	16.2	10.2	4161	1577	4702	2394	0.833	12.0	9.56	822

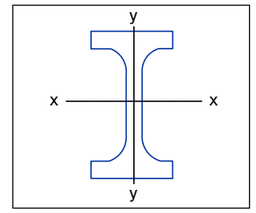
2.3
Universal Beams & Columns (UB & UC)



Imperial Units

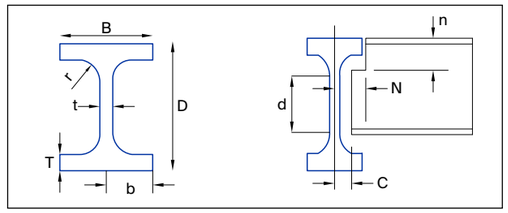
Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W14-14x16 (356x406)	176	261.9	386.6	397.5	33.3	21.1	15.2	289.7	334	5.97	13.7	13	198	48
	193	287.1	393.6	399.0	36.5	22.6	15.2	290.2	366	5.47	12.8	13	198	52
	211	314	399.3	401.3	39.6	24.9	15.2	289.7	400	5.06	11.6	14	198	55
	219	326	403.1	402.0	41.2	25.5	15.2	290.3	415	4.88	11.4	15	198	56
	228	339.9	406.4	403.0	42.9	26.6	15.2	290.2	433	4.70	10.9	15	198	58
	233	346.7	407.4	403.6	43.7	27.2	15.2	289.6	442	4.62	10.7	16	198	59
	257	382.5	416.1	406.3	48.0	29.8	15.2	289.7	488	4.23	9.71	17	198	63
	264	393	419.0	407.0	49.2	30.6	15.2	290.2	501	4.14	9.48	17	198	64
	283	421.1	425.2	409.2	52.6	32.8	15.2	289.6	537	3.89	8.84	18	198	68
	311	462.8	434.8	412.2	57.4	35.8	15.2	289.6	590	3.59	8.09	20	198	73
	314	467	436.6	412.2	58.0	35.8	15.2	290.2	595	3.55	8.11	20	198	73
	342	509	445.5	415.5	62.7	39.1	15.2	289.6	649	3.31	7.40	22	198	78
	370	551	455.6	418.5	67.5	42.1	15.2	290.2	702	3.10	6.89	23	198	83
	398	592.3	464.6	421.4	72.3	45.0	15.2	289.7	755	2.92	6.44	24	198	87
	426	633.9	474.6	424.0	77.0	47.6	15.2	290.2	808	2.75	6.10	26	198	92
	455	677.1	483.1	427.6	81.5	51.2	15.2	289.6	863	2.62	5.66	28	198	97
	500	744.1	497.8	432.1	88.9	55.6	15.2	289.6	948	2.43	5.21	30	198	104
	550	818.5	514.1	436.9	97.0	60.5	15.2	289.6	1043	2.25	4.79	32	198	112
	605	900.3	531.4	442.3	105.6	65.9	15.2	289.8	1147	2.09	4.40	35	198	121
	665	989.6	549.7	448.3	114.8	71.9	15.2	289.7	1261	1.95	4.03	38	198	130
	730	1086	569.5	454.4	124.7	78.0	15.2	289.7	1385	1.82	3.72	41	198	140
W16-16x5 1/2 (406x140)	26	39.0	398.0	141.8	8.6	6.4	10.2	360.4	49.7	8.24	56.3	5	78	19
	31	46.0	403.2	142.2	11.2	6.8	10.2	360.4	58.6	6.35	53.0	5	78	21
W16-16x7 (406x178)	36	54.1	402.6	177.7	10.9	7.7	10.2	360.4	69.0	8.15	46.8	6	95	21
	40	60.1	406.4	177.9	12.8	7.9	10.2	360.4	76.5	6.95	45.6	6	95	23
	45	67.1	409.4	178.8	14.3	8.8	10.2	360.4	85.5	6.25	41.0	6	95	25
	50	74.2	412.8	179.5	16.0	9.5	10.2	360.4	94.5	5.61	37.9	7	95	26
	57	84.8	417.3	180.8	18.2	10.9	10.2	360.6	108	4.98	33.0	7	95	28
W16-16x10 1/4 (406x260)	67	99.7	414.8	260.0	16.9	10.0	10.2	360.6	127	7.70	36.0	7	135	27
	77	114.6	419.6	261.5	19.3	11.6	10.2	360.6	146	6.77	31.2	8	135	30
	89	132.4	425.4	263.3	22.2	13.3	10.2	360.6	169	5.92	27.0	9	135	32
	100	148.8	431.0	264.8	25.0	14.9	10.2	360.6	190	5.29	24.3	9	135	35
W18-18x6 (457x152)	35	52.3	449.8	152.4	10.9	7.6	10.2	407.6	66.6	6.99	53.6	6	82	21
	40	59.8	454.6	152.9	13.3	8.1	10.2	407.6	76.2	5.75	50.3	6	82	24
	45	67.2	458.0	153.8	15.0	9.0	10.2	407.6	85.6	5.13	45.3	7	82	25

Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)

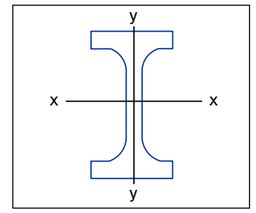


Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
2.29	89040	34860	16.3	10.2	4606	1754	5244	2667	0.833	11.0	10.9	1101
2.31	99880	38680	16.5	10.3	5075	1939	5812	2949	0.835	10.2	12.3	1441
2.33	110500	42720	16.6	10.3	5537	2129	6387	3243	0.834	9.5	13.8	1854
2.34	116400	44660	16.7	10.4	5777	2222	6681	3384	0.836	9.2	14.6	2073
2.35	122500	46850	16.8	10.4	6031	2325	6999	3544	0.836	8.8	15.5	2343
2.35	125100	47930	16.8	10.4	6141	2375	7140	3621	0.836	8.7	15.9	2478
2.37	141500	53750	17.0	10.5	6803	2646	7975	4038	0.837	8.0	18.2	3292
2.38	146600	55370	17.1	10.5	6998	2721	8222	4154	0.837	7.9	18.9	3545
2.40	159800	60150	17.2	10.6	7516	2940	8887	4492	0.837	7.4	20.9	4336
2.42	180100	67130	17.5	10.7	8283	3257	9877	4983	0.838	6.9	23.9	5655
2.42	183000	67830	17.5	10.7	8383	3291	10000	5034	0.839	6.9	24.3	5809
2.45	203800	75180	17.7	10.8	9151	3619	11010	5543	0.840	6.4	27.5	7401
2.47	226900	82670	18.0	10.9	9962	3951	12080	6058	0.841	6.1	31.1	9240
2.50	249800	90380	18.2	10.9	10750	4289	13130	6583	0.842	5.7	34.8	11350
2.52	274800	98130	18.4	11.0	11580	4629	14240	7108	0.843	5.5	38.8	13720
2.55	299400	106600	18.6	11.1	12400	4987	15340	7669	0.843	5.2	43.0	16460
2.59	341900	120000	19.0	11.3	13740	5555	17160	8553	0.844	4.9	50.2	21390
2.63	392400	135500	19.4	11.4	15260	6202	19260	9560	0.846	4.6	58.9	27870
2.67	450600	153100	19.8	11.6	16960	6922	21610	10680	0.848	4.3	69.4	36130
2.72	518100	173400	20.3	11.7	18850	7736	24250	11960	0.849	4.0	82.0	46760
2.77	597000	196300	20.8	11.9	20970	8640	27230	13370	0.851	3.8	97.1	60230
1.33	12510	410	15.9	2.87	629	57.8	724	90.8	0.858	47.5	0.155	10.7
1.34	15690	538	16.4	3.03	778	75.7	888	118	0.871	38.9	0.207	19.0
1.48	18720	1021	16.5	3.85	930	115	1055	178	0.871	38.3	0.392	23.1
1.49	21600	1203	16.8	3.97	1063	135	1199	209	0.880	33.8	0.466	33.3
1.50	24330	1365	16.9	3.99	1189	153	1346	237	0.880	30.5	0.533	46.1
1.51	27310	1546	17.0	4.04	1323	172	1501	267	0.882	27.6	0.608	62.8
1.52	31520	1794	17.1	4.07	1511	198	1724	309	0.881	24.5	0.714	92.2
1.83	39730	4951	17.7	6.25	1915	381	2128	581	0.888	25.4	1.96	99.4
1.84	46110	5758	17.8	6.28	2198	440	2457	673	0.887	22.4	2.31	149
1.86	54060	6768	17.9	6.33	2541	514	2859	788	0.887	19.7	2.75	227
1.87	61830	7754	18.0	6.39	2869	586	3246	899	0.887	17.7	3.20	322
1.48	21370	645	17.9	3.11	950	84.7	1096	133	0.859	43.9	0.311	21.4
1.49	25500	795	18.3	3.23	1122	104	1287	163	0.868	37.5	0.387	33.8
1.50	28930	913	18.4	3.27	1263	119	1453	187	0.869	33.6	0.448	47.7

2.3
Universal Beams & Columns (UB & UC)



Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)

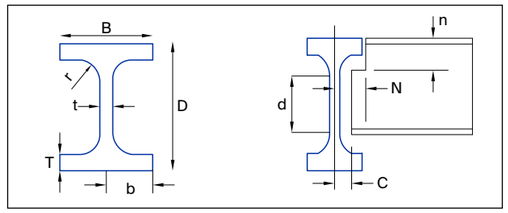


Imperial Units

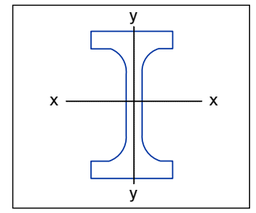
Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W18-18x6 (457x152)	46	68.5	458.7	153.9	15.3	9.1	10.2	407.7	87.1	5.03	44.6	7	82	26
	50	74.2	462.0	154.4	17.0	9.6	10.2	407.6	94.5	4.54	42.5	7	82	27
	55	82.1	465.8	155.3	18.9	10.5	10.2	407.6	105	4.11	38.8	7	82	29
W18-18x7 1/2 (457x191)	45	67.1	453.4	189.9	12.7	8.5	10.2	407.6	85.5	7.48	48.0	6	101	23
	50	74.3	457.0	190.4	14.5	9.0	10.2	407.6	94.6	6.57	45.3	7	101	25
	55	82.0	460.0	191.3	16.0	9.9	10.2	407.6	104	5.98	41.2	7	101	26
	60	89.3	463.4	191.9	17.7	10.5	10.2	407.6	114	5.42	38.8	7	101	28
	65	96.7	466.1	192.8	19.1	11.4	10.2	407.6	123	5.06	35.7	8	101	29
	66	98.3	467.2	192.8	19.6	11.4	10.2	407.6	125	4.92	35.8	8	101	30
	71	105.7	469.1	193.9	20.6	12.6	10.2	407.6	134	4.71	32.4	8	101	31
W18-18x11 (457x279)	76	113.1	462.5	280.3	17.3	10.8	10.2	407.6	144	8.12	37.7	7	145	27
	86	128	467.1	281.7	19.6	12.1	10.2	407.6	163	7.20	33.7	8	145	30
	97	144.4	472.2	283.1	22.1	13.6	10.2	407.6	184	6.40	30.0	9	145	32
	106	157.7	475.7	284.5	23.9	15.0	10.2	407.5	201	5.96	27.2	9	145	34
	119	177.1	481.8	286.1	26.9	16.6	10.2	407.6	226	5.31	24.5	10	145	37
	130	193.5	489.0	283.5	30.5	17.0	10.2	407.6	247	4.65	24.0	11	143	41
	143	212.8	495.0	285.0	33.5	18.5	10.2	407.5	271	4.25	22.0	11	143	44
W21-21x6 1/2 (533x165)	44	65.5	524.8	165.1	11.4	8.9	12.7	476.6	84	7.24	53.6	6	88	24
	50	74.4	529.1	165.9	13.5	9.7	12.7	476.7	95	6.14	49.4	7	88	26
	57	84.8	534.9	166.5	16.5	10.3	12.7	476.5	108	5.04	46.3	7	88	29
W21-21x8 1/4 (533x210)	55	82.2	528.3	208.8	13.2	9.6	12.7	476.5	105	7.91	49.6	7	110	26
	62	92.1	533.1	209.3	15.6	10.1	12.7	476.5	117	6.71	47.2	7	110	28
	68	101	536.7	210.0	17.4	10.8	12.7	476.5	129	6.03	44.1	7	110	30
	73	109	539.5	210.8	18.8	11.6	12.7	476.5	139	5.61	41.1	8	110	32
	82	122	544.5	211.9	21.3	12.7	12.7	476.5	155	4.97	37.5	8	110	34
	83	123.5	544.3	212.2	21.2	13.1	12.7	476.5	157	5.00	36.4	9	110	34
	93	138.4	549.1	213.9	23.6	14.7	12.7	476.5	176	4.53	32.3	9	110	36
W21-21x12 1/4 (533x312)	101	150.3	542.5	312.2	20.3	12.7	12.7	476.5	192	7.68	37.5	8	160	33
	111	165.2	546.4	313.4	22.2	14.0	12.7	476.6	211	7.05	34.1	9	160	35
	122	181.6	550.7	314.7	24.4	15.2	12.7	476.5	231	6.45	31.3	10	160	37
	132	196.4	554.5	316.0	26.3	16.5	12.7	476.5	250	6.01	28.9	10	160	39
	147	218.8	560.3	317.8	29.2	18.3	12.7	476.5	279	5.44	26.1	11	160	42
	166	247	571.0	315.5	34.5	19.1	12.7	476.5	315	4.57	25.0	12	158	47
	182	270.8	577.1	317.8	37.6	21.1	12.7	476.5	346	4.23	22.6	13	158	50

Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³	u	x	H	J
1.50	29530	933	18.4	3.27	1288	121	1482	191	0.869	33.0	0.458	50.4
1.50	32670	1047	18.6	3.33	1414	136	1627	213	0.873	30.1	0.518	65.9
1.51	36590	1185	18.7	3.37	1571	153	1811	240	0.873	27.4	0.591	89.2
1.63	29380	1452	18.5	4.12	1296	153	1471	237	0.872	37.9	0.705	37.1
1.64	33320	1671	18.8	4.20	1458	176	1653	272	0.877	33.9	0.818	51.8
1.65	37050	1871	18.8	4.23	1611	196	1831	304	0.877	30.9	0.922	69.2
1.66	41020	2089	19.0	4.29	1770	218	2014	338	0.880	28.3	1.04	90.7
1.66	44590	2281	19.0	4.30	1913	237	2184	369	0.879	26.3	1.14	114
1.67	45730	2347	19.1	4.33	1957	243	2232	379	0.881	25.7	1.18	121
1.67	48760	2507	19.0	4.32	2079	259	2383	404	0.877	24.4	1.26	145
2.01	55460	6344	19.6	6.64	2398	453	2669	692	0.885	27.9	3.14	118
2.02	63520	7295	19.7	6.69	2720	518	3039	792	0.886	24.8	3.65	170
2.03	72710	8367	19.9	6.74	3079	591	3457	906	0.886	22.1	4.24	244
2.04	79600	9178	19.9	6.76	3347	645	3775	991	0.884	20.5	4.68	311
2.06	91040	10530	20.1	6.82	3779	736	4284	1132	0.885	18.4	5.44	442
2.06	102500	11590	20.4	6.86	4192	818	4761	1257	0.891	16.6	6.09	603
2.08	114400	12960	20.5	6.91	4624	910	5278	1400	0.891	15.2	6.90	800
1.67	35030	859	20.5	3.20	1335	104	1561	166	0.847	47.0	0.566	31.9
1.68	40810	1032	20.8	3.30	1543	124	1797	199	0.853	41.4	0.686	47.1
1.69	48640	1276	21.2	3.44	1819	153	2107	243	0.862	35.5	0.857	73.8
1.85	47540	2008	21.3	4.38	1800	192	2059	300	0.864	41.6	1.33	51.5
1.86	55230	2389	21.7	4.51	2072	228	2360	356	0.872	36.5	1.60	75.7
1.87	61520	2692	21.9	4.57	2292	256	2612	399	0.874	33.2	1.82	101
1.88	66820	2943	21.9	4.60	2477	279	2828	436	0.875	30.9	1.99	126
1.89	76040	3388	22.1	4.67	2793	320	3196	500	0.877	27.6	2.32	178
1.89	76240	3389	22.0	4.65	2801	319	3212	500	0.874	27.6	2.32	181
1.90	86170	3868	22.1	4.68	3139	362	3617	569	0.873	24.9	2.67	251
2.29	100800	10320	22.9	7.33	3715	661	4147	1012	0.885	27.8	7.03	217
2.30	111300	11410	23.0	7.36	4074	728	4565	1117	0.885	25.6	7.84	284
2.31	123300	12680	23.1	7.40	4476	806	5032	1238	0.885	23.4	8.78	373
2.32	134200	13850	23.2	7.44	4842	876	5462	1348	0.884	21.8	9.66	469
2.33	151200	15650	23.3	7.49	5396	985	6116	1519	0.884	19.8	11.0	643
2.34	178000	18110	23.8	7.58	6233	1148	7080	1766	0.891	17.2	13.0	983
2.36	197200	20150	23.9	7.63	6834	1268	7807	1956	0.890	15.9	14.7	1278

2.3
Universal Beams & Columns (UB & UC)



Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)

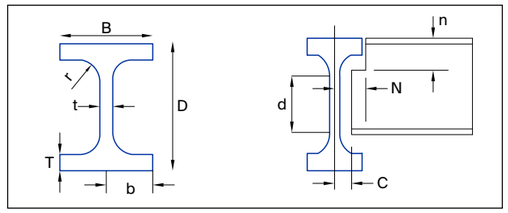


Imperial Units

Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W21-21x12 1/4 (533x312)	201	300	585.0	319.0	41.4	23.1	12.7	476.8	382	3.85	20.6	14	158	54
	223	331	593.0	322.0	45.5	25.4	12.7	476.6	422	3.54	18.8	15	158	58
	248	370	603.0	324.0	50.5	26.2	12.7	476.6	460	3.21	18.2	15	159	63
	275	409	613.0	327.0	55.6	31.0	12.7	476.4	521	2.94	15.4	18	158	68
W24-24x7 (610x178)	55	81.9	598.7	177.9	12.8	10.0	12.7	547.6	105	6.93	54.6	7	94	26
	61	91.0	602.5	178.4	15.0	10.6	12.7	547.1	116	5.95	51.6	7	94	28
	62	92.3	603.0	178.8	15.0	10.9	12.7	547.6	118	5.96	50.1	7	94	28
W24-24x9 (610x229)	68	101.2	602.6	227.6	14.8	10.5	12.7	547.6	129	7.69	52.2	7	119	28
	76	113	607.6	228.2	17.3	11.1	12.7	547.6	144	6.60	49.3	8	119	30
	84	125.1	612.2	229.0	19.3	11.9	12.7	548.2	158	5.93	46.1	8	119	32
	94	139.9	617.2	230.2	22.1	13.1	12.7	547.6	178	5.21	41.8	9	119	35
	103	153	623.1	228.6	24.9	14.0	12.7	547.9	195	4.59	39.1	9	117	38
	114	171	628.9	229.9	27.9	15.5	12.7	547.7	218	4.12	35.3	10	117	41
	128	191	635.0	231.5	30.9	17.0	12.7	547.8	242	3.75	32.2	11	117	44
	146	217	643.1	234.1	35.1	19.6	12.7	547.5	278	3.33	27.9	12	117	48
	163	243	651.0	235.9	39.1	21.6	12.7	547.4	310	3.02	25.3	13	117	52
	181	270	658.6	238.5	42.9	23.9	12.7	547.4	343	2.78	22.9	14	117	56
	198	295	667.0	240.5	47.0	25.9	12.7	547.6	376	2.56	21.1	15	117	60
	218	325	675.1	243.1	51.0	28.4	12.7	547.7	412	2.38	19.3	16	117	64
	239	356	685.0	245.6	55.9	31.0	12.7	547.8	454	2.20	17.7	18	117	69
W24-24x12 (610x305)	100	149.1	612.4	304.8	19.7	11.8	16.5	540.0	190	7.74	45.8	8	157	36
	120	179	620.2	307.1	23.6	14.1	16.5	540.0	228	6.51	38.3	9	157	40
	160	238.1	635.8	311.4	31.4	18.4	16.5	540.0	303	4.96	29.3	11	157	48
W24-24x12 3/4 (610x324)	104	154.8	611.1	323.8	19.1	12.7	12.7	547.6	198	8.50	43.1	8	166	32
	117	174.1	616.2	325.1	21.6	14.0	12.7	547.6	222	7.53	39.2	9	166	34
	131	194.9	621.8	326.5	24.4	15.4	12.7	547.6	249	6.70	35.6	10	166	37
	146	217.3	628.4	327.7	27.7	16.5	12.7	547.6	277	5.92	33.2	10	166	40
	162	241.1	635.0	329.1	31.0	17.9	12.7	547.6	308	5.31	30.6	11	166	44
	176	261.9	641.1	327.4	34.0	19.1	12.7	547.6	333	4.81	28.7	12	164	47
	192	285.7	646.9	328.9	37.1	20.6	12.7	547.3	363	4.44	26.6	12	164	50
	207	308	653.0	330.5	39.9	22.1	12.7	547.8	392	4.14	24.8	13	164	53
	229	341	661.0	333.0	43.9	24.4	16.5	540.2	435	3.79	22.1	14	164	60
	250	372	669.0	335.0	48.0	26.4	16.5	540.0	475	3.49	20.5	15	164	65
	279	415	679.0	338.0	53.1	29.5	16.5	539.8	530	3.18	18.3	17	164	70
	306	455	689.0	340.0	57.9	32.0	16.5	540.2	579	2.94	16.9	18	164	74

Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³	u	x	H	J
2.38	220700	22450	24.1	7.67	7547	1408	8670	2175	0.890	14.6	16.6	1700
2.40	247700	25390	24.2	7.76	8355	1577	9656	2442	0.890	13.4	19.0	2263
2.43	278900	28710	24.6	7.90	9250	1772	10730	2739	0.894	12.3	21.9	2996
2.45	316900	32530	24.7	7.91	10340	1990	12120	3096	0.890	11.2	25.3	4131
1.87	56020	1210	23.2	3.40	1871	136	2200	219	0.843	48.3	1.04	49.2
1.88	63880	1426	23.5	3.51	2120	160	2480	256	0.850	43.2	1.23	68.6
1.88	64580	1436	23.4	3.49	2142	161	2512	258	0.848	42.8	1.24	71.0
2.07	75780	2915	24.2	4.76	2515	256	2881	400	0.864	43.1	2.52	77.0
2.08	87320	3434	24.6	4.88	2874	301	3281	469	0.870	38.0	2.99	111
2.09	97540	3872	24.8	4.95	3187	338	3639	528	0.872	34.5	3.40	149
2.11	111800	4505	25.0	5.03	3622	391	4142	611	0.875	30.6	3.99	216
2.11	125000	4972	25.3	5.04	4012	435	4595	680	0.878	27.6	4.45	295
2.12	141400	5670	25.4	5.09	4495	493	5167	773	0.878	24.8	5.12	411
2.14	158400	6415	25.6	5.15	4990	554	5757	871	0.879	22.6	5.85	554
2.16	183900	7544	25.7	5.21	5718	644	6643	1019	0.877	20.0	6.97	819
2.18	207900	8606	25.9	5.27	6386	730	7455	1157	0.878	18.2	8.06	1123
2.20	232800	9768	26.1	5.34	7069	819	8299	1304	0.877	16.7	9.26	1495
2.22	259400	10980	26.3	5.41	7778	913	9173	1458	0.878	15.4	10.6	1952
2.24	287700	12330	26.4	5.47	8522	1014	10110	1625	0.877	14.3	12.0	2517
2.27	322200	13950	26.6	5.55	9406	1136	11220	1826	0.877	13.2	13.8	3309
2.39	125900	9308	25.7	7.00	4111	611	4594	937	0.886	32.7	8.17	200
2.41	153000	11410	25.9	7.07	4935	743	5547	1144	0.886	27.7	10.2	340
2.45	209500	15840	26.3	7.23	6589	1017	7486	1574	0.886	21.3	14.5	785
2.47	129200	10790	25.6	7.39	4227	666	4734	1023	0.880	33.6	9.46	196
2.48	147200	12380	25.8	7.47	4776	762	5360	1170	0.882	30.0	10.9	280
2.50	167300	14160	25.9	7.55	5383	867	6057	1335	0.883	26.8	12.6	395
2.51	190800	16260	26.2	7.66	6074	993	6845	1527	0.886	23.9	14.7	559
2.53	215400	18440	26.4	7.74	6784	1121	7670	1726	0.888	21.6	16.8	770
2.53	236600	19950	26.6	7.73	7380	1218	8369	1878	0.890	19.9	18.4	995
2.55	260400	22030	26.8	7.79	8050	1340	9163	2068	0.890	18.4	20.5	1280
2.56	283900	24050	26.9	7.84	8695	1455	9936	2250	0.890	17.2	22.6	1593
2.58	319000	27090	27.1	7.90	9652	1627	11090	2523	0.890	15.7	25.8	2162
2.60	353900	30170	27.3	7.97	10580	1801	12220	2797	0.890	14.5	29.1	2809
2.62	400500	34310	27.5	8.04	11800	2030	13720	3162	0.889	13.2	33.6	3824
2.65	445200	38100	27.7	8.11	12920	2241	15120	3498	0.889	12.2	37.9	4940

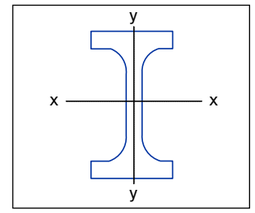
2.3
Universal Beams & Columns (UB & UC)



Imperial Units

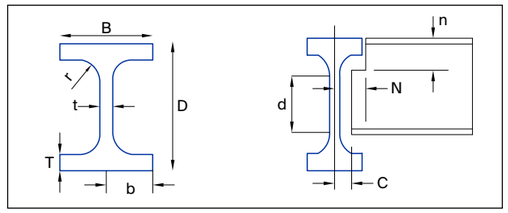
Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W27-27x10 (686x254)	84	125.2	677.9	253.0	16.2	11.7	15.2	615.1	159	7.81	52.6	8	131	31
	94	140.1	683.5	253.7	19.0	12.4	15.2	615.1	178	6.68	49.6	8	131	34
	102	152.4	687.5	254.5	21.0	13.2	15.2	615.1	194	6.06	46.6	9	131	36
	114	170.2	692.9	255.8	23.7	14.5	15.2	615.1	217	5.40	42.4	9	131	39
	129	192	701.8	254.3	27.9	15.5	15.2	615.6	244	4.56	39.7	10	129	43
	143	213	707.9	255.8	31.0	17.0	15.2	615.5	270	4.13	36.2	11	129	46
	159	237	714.0	257.8	34.0	19.1	15.2	615.6	301	3.79	32.2	12	129	49
	182	271	723.9	260.2	39.1	21.6	15.2	615.3	345	3.33	28.5	13	129	54
	201	299	732.0	262.8	42.9	23.9	15.2	615.8	382	3.06	25.8	14	129	58
	221	329	739.9	264.8	47.0	25.9	15.2	615.5	418	2.82	23.8	15	129	62
	247	368	749.8	267.7	52.1	28.9	15.2	615.2	468	2.57	21.3	16	129	67
	271	404	760.0	270.3	56.9	31.5	15.2	615.8	513	2.38	19.5	18	129	72
	302	449	771.9	273.8	63.0	35.1	15.2	615.5	574	2.17	17.5	20	129	78
W27-27x14 (686x356)	146	217.3	695.5	354.7	24.8	15.4	15.2	615.6	277	7.16	40.1	10	180	40
	161	239.6	700.8	356.1	27.4	16.8	15.2	615.5	306	6.49	36.7	10	180	43
	178	264.9	706.4	357.8	30.2	18.4	15.2	615.5	337	5.92	33.4	11	180	45
	194	288.7	714.0	356.4	34.0	19.1	15.2	615.5	368	5.24	32.3	12	179	49
	217	322.9	722.1	358.5	38.1	21.1	15.2	615.5	411	4.70	29.2	13	179	53
	235	349.7	728.0	360.4	40.9	23.1	15.2	615.8	446	4.41	26.6	14	179	56
	258	383.9	736.1	362.5	45.0	24.9	15.2	615.8	489	4.03	24.7	14	179	60
	281	418.2	744.0	364.5	49.0	26.9	15.2	615.6	533	3.72	22.9	15	179	64
	307	456.9	752.1	366.9	53.1	29.5	15.2	615.5	582	3.46	20.9	17	179	68
	336	500	762.0	369.0	57.9	32.0	15.2	615.8	636	3.19	19.2	18	179	73
W30-30x10 1/2 (762x267)	90	133.9	750.0	264.4	15.5	12.0	16.5	686.0	171	8.53	57.2	8	136	32
	99	146.9	754.0	265.2	17.5	12.8	16.5	686.0	187	7.58	53.6	8	136	34
	108	160.7	757.7	266.1	19.3	13.8	16.5	686.1	205	6.89	49.6	9	136	36
	116	173	762.2	266.7	21.6	14.3	16.5	686.0	220	6.17	48.0	9	136	38
	124	184.5	766.3	267.1	23.6	14.9	16.5	686.1	235	5.65	46.2	9	136	40
	132	196.8	769.8	268.0	25.4	15.6	16.5	686.0	251	5.28	44.0	10	136	42
	148	220	779.0	266.2	30.0	16.5	16.5	686.0	281	4.44	41.6	10	135	47
	165	246	785.1	268.1	33.0	18.5	16.5	686.1	312	4.06	37.1	11	135	50
	185	275	793.0	270.1	37.1	20.6	16.5	685.8	351	3.64	33.3	12	135	54
	207	308	801.1	272.7	40.9	23.1	16.5	686.3	392	3.33	29.7	14	135	57
	226	337	809.0	274.6	44.9	24.9	16.5	686.2	428	3.06	27.6	14	135	61
	246	366	817.1	276.6	48.0	26.9	16.5	688.1	462	2.88	25.6	15	135	65

Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)



Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
2.32	118000	4383	27.2	5.24	3481	347	3994	542	0.862	43.9	4.80	116
2.33	136300	5184	27.6	5.39	3987	409	4558	638	0.868	38.7	5.72	169
2.34	150400	5784	27.8	5.46	4374	455	5000	710	0.871	35.5	6.42	220
2.35	170300	6631	28.0	5.53	4916	518	5631	811	0.872	31.8	7.42	308
2.36	198000	7670	28.5	5.61	5644	603	6462	943	0.878	27.7	8.71	462
2.38	222000	8678	28.7	5.67	6272	678	7204	1063	0.878	25.2	9.94	627
2.39	247800	9750	28.7	5.69	6940	756	8016	1191	0.877	23.0	11.3	840
2.42	289300	11540	29.0	5.78	7993	887	9282	1402	0.878	20.3	13.5	1260
2.44	323800	13060	29.1	5.85	8847	994	10330	1577	0.877	18.6	15.5	1675
2.46	359400	14640	29.3	5.92	9715	1106	11390	1759	0.877	17.2	17.6	2186
2.49	406900	16800	29.5	5.99	10850	1255	12810	2005	0.877	15.6	20.4	2993
2.51	453800	18910	29.7	6.07	11940	1399	14170	2243	0.877	14.4	23.4	3897
2.54	515400	21790	30.0	6.16	13350	1592	15950	2565	0.876	13.2	27.4	5322
2.75	234200	18440	29.1	8.16	6735	1040	7558	1598	0.884	29.7	20.7	454
2.77	261200	20670	29.2	8.22	7455	1161	8389	1787	0.885	27.0	23.4	610
2.78	290800	23120	29.4	8.28	8233	1292	9297	1992	0.885	24.6	26.4	813
2.79	325500	25720	29.8	8.36	9118	1444	10300	2223	0.889	22.3	29.7	1100
2.81	369200	29310	30.0	8.44	10230	1635	11610	2523	0.890	20.1	34.3	1532
2.83	402300	31970	30.0	8.47	11050	1774	12600	2745	0.888	18.8	37.7	1914
2.85	447800	35780	30.3	8.56	12170	1974	13930	3057	0.889	17.2	42.7	2518
2.87	494700	39680	30.5	8.63	13300	2177	15290	3377	0.890	15.9	47.9	3250
2.89	545000	43850	30.6	8.68	14990	2390	16750	3717	0.889	14.8	53.6	4152
2.91	604800	48670	30.8	8.75	15870	2638	18450	4111	0.889	13.7	60.3	5373
2.51	150700	4788	29.7	5.30	4018	362	4644	570	0.854	49.8	6.46	119
2.51	168500	5456	30.0	5.40	4470	411	5156	647	0.858	45.2	7.40	159
2.52	185900	6080	30.1	5.45	4907	457	5665	720	0.859	41.5	8.29	208
2.53	205300	6850	30.5	5.58	5387	514	6198	807	0.864	38.1	9.39	267
2.54	223100	7525	30.8	5.65	5821	563	6689	885	0.867	35.4	10.4	333
2.55	240000	8175	30.9	5.71	6234	610	7167	959	0.869	33.2	11.3	404
2.56	278200	9463	31.5	5.81	7142	711	8197	1115	0.875	28.9	13.3	605
2.58	310700	10640	31.5	5.84	7914	794	9129	1251	0.874	26.4	15.0	815
2.60	353200	12240	31.7	5.91	8909	906	10320	1433	0.874	23.7	17.5	1147
2.62	397200	13900	31.8	5.96	9916	1020	11550	1620	0.873	21.6	20.1	1555
2.64	440500	15600	32.1	6.04	10890	1136	12720	1808	0.874	19.9	22.8	2029
2.66	480200	17050	32.2	6.08	11750	1233	13790	1971	0.873	18.7	25.2	2500

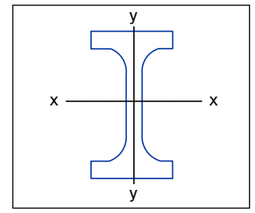
2.3
Universal Beams & Columns (UB & UC)



Imperial Units

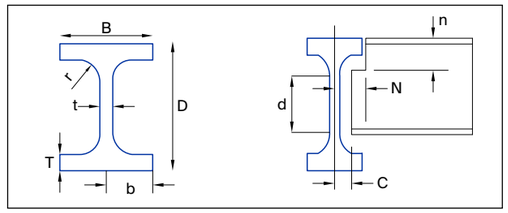
Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W30-30x10 1/2 (762x267)	269	401	825.0	279.1	53.1	29.5	16.5	685.8	511	2.63	23.2	17	135	70
	295	439	835.2	281.7	57.9	32.0	16.5	686.4	559	2.43	21.5	18	135	74
W30-30x15 (762x381)	173	257.5	773.2	380.6	27.1	16.6	16.5	686.1	328	7.04	41.2	10	192	44
	191	284.2	779.3	382.0	30.1	18.0	16.5	686.1	362	6.35	38.1	11	192	47
	211	314	785.9	383.7	33.4	19.7	16.5	686.1	400	5.74	34.9	12	192	50
	235	349.7	795.0	382.4	38.1	21.1	16.5	685.8	445	5.02	32.5	13	191	55
	261	388.4	802.9	384.9	41.9	23.6	16.5	686.1	495	4.59	29.0	14	191	58
	292	434.5	813.1	387.5	47.0	25.9	16.5	686.1	553	4.12	26.5	15	191	63
	326	485.1	823.0	390.4	52.1	29.0	16.5	685.9	617	3.75	23.7	16	191	69
	357	531	833.0	393.0	56.9	31.5	20.0	679.2	677	3.45	21.6	18	191	77
W33-33x11 1/2 (838x292)	118	175.9	834.9	291.7	18.8	14.0	17.8	761.7	224	7.76	54.4	9	149	37
	130	193.8	840.7	292.4	21.7	14.7	17.8	761.7	247	6.74	51.8	9	149	40
	141	209.8	845.8	293.0	24.4	15.4	17.8	761.4	268	6.01	49.5	10	149	42
	152	226.5	850.9	293.8	26.8	16.1	17.8	761.7	289	5.48	47.3	10	149	45
	169	252	859.0	292.1	31.0	17.0	17.8	761.4	319	4.71	44.8	11	148	49
	187	278	865.1	294.1	34.0	19.1	17.8	761.5	355	4.33	39.9	12	148	52
	204	304	871.2	295.7	37.1	20.6	17.8	761.4	386	3.99	37.0	12	148	55
	219	326	877.1	297.2	39.9	22.1	17.8	761.7	416	3.72	34.5	13	148	58
	243	362	885.2	299.7	43.9	24.4	17.8	761.8	460	3.41	31.2	14	148	62
	271	404	895.1	302.1	49.0	26.9	17.8	761.5	513	3.08	28.3	15	148	67
	301	449	905.0	305.2	54.1	30.0	17.8	761.2	572	2.82	25.4	17	148	72
	332	494	915.2	308.1	58.9	33.0	17.8	761.8	629	2.62	23.1	19	148	77
	361	537	925.1	310.6	64.0	35.6	17.8	761.5	684	2.43	21.4	20	148	82
W33-33x15 3/4 (838x400)	201	299.1	855.5	399.9	29.2	18.2	17.8	761.5	381	6.85	41.9	11	201	47
	221	328.9	861.8	401.4	32.4	19.7	17.8	761.4	420	6.20	38.7	12	201	50
	241	358.6	868.2	402.8	35.6	21.1	17.8	761.5	457	5.66	36.1	13	201	53
	263	391.4	877.1	401.4	39.9	22.1	17.8	761.7	499	5.03	34.5	13	200	58
	291	433.1	884.9	404.0	43.9	24.4	17.8	761.4	552	4.60	31.2	14	200	62
	318	473.2	893.1	406.0	48.0	26.4	17.8	761.5	603	4.23	28.8	15	200	66
	354	526.8	903.0	408.9	53.1	29.5	17.8	761.2	672	3.85	25.8	17	200	71
	387	575.9	913.1	411.5	57.9	32.0	17.8	761.7	734	3.55	23.8	18	200	76
	424	631	923.0	414.0	63.0	35.1	20.0	757.0	805	3.29	21.6	20	199	83
	468	697	935.0	418.0	69.1	38.6	20.0	756.8	889	3.02	19.6	21	200	89

Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)

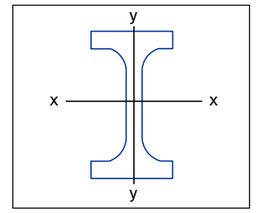


Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
2.68	536500	19400	32.4	6.16	13010	1390	15330	2229	0.874	17.0	28.9	3361
2.70	595900	21780	32.7	6.24	14270	1546	16900	2486	0.874	15.8	32.9	4345
3.01	341200	24890	32.3	8.71	8827	1308	9916	2012	0.883	30.3	34.6	635
3.02	381700	28000	32.5	8.80	9796	1466	11030	2258	0.885	27.5	39.3	859
3.04	427000	31500	32.7	8.87	10870	1642	12270	2531	0.886	25.0	44.6	1161
3.05	485900	35570	33.0	8.94	12220	1860	13830	2869	0.889	22.2	50.9	1651
3.07	543700	39920	33.1	8.98	13540	2074	15410	3208	0.888	20.3	57.8	2218
3.10	618300	45680	33.4	9.09	15210	2358	17380	3653	0.889	18.3	67.0	3090
3.12	697600	51790	33.6	9.16	16950	2653	19500	4123	0.888	16.7	77.0	4224
3.14	776600	57770	33.9	9.24	18650	2940	21550	4579	0.888	15.3	87.0	5563
2.78	246000	7800	33.1	5.90	5893	535	6808	842	0.856	46.5	13.0	221
2.79	279200	9067	33.6	6.06	6642	620	7640	974	0.862	41.6	15.2	306
2.80	310200	10250	34.0	6.18	7334	700	8416	1097	0.867	37.9	17.3	404
2.81	339700	11360	34.3	6.27	7985	773	9155	1212	0.870	35.0	19.3	514
2.82	386500	12920	34.8	6.36	8999	884	10310	1383	0.875	30.9	22.1	735
2.84	430400	14470	34.8	6.38	9950	984	11450	1547	0.874	28.3	25.0	983
2.85	473000	16050	35.0	6.45	10860	1086	12530	1710	0.874	26.1	27.9	1265
2.87	513500	17540	35.1	6.49	11710	1180	13550	1864	0.874	24.4	30.7	1569
2.89	573400	19800	35.3	6.56	12960	1321	15060	2095	0.874	22.3	35.0	2095
2.91	648200	22660	35.5	6.64	14480	1500	16910	2385	0.875	20.2	40.5	2884
2.94	729200	25820	35.7	6.72	16120	1692	18920	2704	0.874	18.4	46.7	3910
2.97	810000	28960	35.9	6.79	17700	1880	20890	3018	0.873	17.0	53.1	5088
2.99	892800	32280	36.1	6.87	19300	2078	22880	3346	0.873	15.8	59.8	6494
3.24	479800	31180	35.5	9.05	11220	1559	12640	2405	0.881	31.2	53.2	855
3.26	534700	34970	35.7	9.13	12410	1742	14020	2690	0.882	28.4	60.1	1146
3.27	590000	38800	35.9	9.21	13590	1927	15380	2977	0.884	26.1	67.3	1491
3.29	659000	43070	36.3	9.29	15030	2146	17020	3314	0.888	23.6	75.5	2008
3.31	735400	48390	36.5	9.36	16620	2396	18910	3709	0.887	21.6	85.6	2687
3.33	812500	53680	36.7	9.43	18200	2644	20780	4101	0.888	19.9	95.8	3488
3.35	913500	60680	36.9	9.50	20330	2968	23230	4616	0.887	18.1	110	4737
3.38	1012000	67480	37.1	9.59	22170	3280	25570	5113	0.887	16.7	123	6135
3.40	1120000	74810	37.3	9.64	24260	3614	28140	5652	0.886	15.5	138	7983
3.43	1253000	84520	37.6	9.75	26810	4044	31270	6342	0.886	14.2	158	10569

2.3
Universal Beams & Columns (UB & UC)



Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)

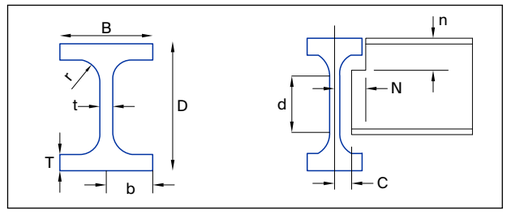


Imperial Units

Designation			Depth of section	Width of section	Thickness		Root radius	Depth between fillets	Area of section	Ratios for local buckling		Dimensions for detailing		
					Flange	Web				Flange	Web	End clearance	Notch	
Size	Mass per metre		D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W36-36x12 (914x305)	135	200.9	903.0	303.3	20.2	15.1	19.1	824.4	256	7.51	54.6	10	154	39
	150	224.2	910.4	304.1	23.9	15.9	19.1	824.4	286	6.36	51.8	10	154	43
	160	238.1	914.7	304.8	25.9	16.5	19.0	824.9	304	5.88	50.0	10	154	45
	170	253.4	918.4	305.5	27.9	17.3	19.1	824.4	323	5.47	47.7	11	154	47
	182	270.8	922.8	306.7	30.0	18.4	19.0	824.9	346	5.12	44.8	11	154	49
	194	289.1	926.6	307.7	32.0	19.5	19.1	824.4	368	4.81	42.3	12	154	51
	210	312.5	931.9	309.4	34.5	21.1	19.0	824.8	399	4.48	39.1	13	154	54
	232	345	942.9	307.8	39.9	22.1	19.1	824.9	440	3.86	37.3	13	153	59
	256	381	950.7	310.3	43.9	24.4	19.1	824.7	486	3.53	33.8	14	153	63
	286	426	960.9	312.8	48.0	26.9	19.1	826.7	536	3.26	30.7	15	153	67
	318	474	970.8	315.7	54.1	29.9	19.1	824.4	603	2.92	27.6	17	153	73
	350	521	980.7	318.8	58.9	33.0	19.1	824.7	663	2.71	25.0	19	153	78
	387	576	992.9	321.8	65.0	36.1	19.1	824.7	733	2.48	22.8	20	153	84
W36-36x16 1/2 (914x419)	230	343.3	911.8	418.5	32.0	19.4	24.1	799.6	437	6.54	41.2	12	210	56
	245	364.6	916.4	419.4	34.3	20.3	24.1	799.6	465	6.12	39.4	12	210	58
	260	388	921.0	420.5	36.6	21.4	24.1	799.6	494	5.74	37.4	13	210	61
	280	416.7	927.6	421.5	39.9	22.5	24.1	799.6	532	5.28	35.6	13	210	64
	300	446.4	933.2	423.0	42.7	24.0	24.1	799.7	569	4.96	33.3	14	210	67
	328	488.1	942.1	422.4	47.0	25.9	24.1	799.9	622	4.49	30.9	15	208	71
	359	534.2	950.0	424.9	51.1	28.5	24.1	799.7	680	4.16	28.1	16	208	75
	393	584.8	960.1	427.5	55.9	31.0	24.1	800.1	746	3.83	25.8	17	208	80
W40-40x12 (1016x305)	149	222	970.3	300.0	21.1	16.0	30.0	868.1	283	7.11	54.3	10	152	51
	167	249	980.2	300.0	26.0	16.5	30.0	868.2	317	5.77	52.6	10	152	56
	183	272	990.1	300.0	31.0	16.5	30.0	868.1	347	4.84	52.6	10	152	61
	211	314	1000.0	300.0	35.9	19.1	30.0	868.2	400	4.18	45.5	12	150	66
	235	349	1008.1	302.0	40.0	21.1	30.0	868.1	445	3.78	41.1	13	150	70
	264	393	1016.0	303.0	43.9	24.4	30.0	868.2	500	3.45	35.6	14	149	74
	294	438	1025.9	305.4	49.0	26.9	30.0	867.9	557	3.12	32.3	15	149	79
	327	487	1036.1	308.5	54.1	30.0	30.0	867.9	620	2.85	28.9	17	149	84
	359	535	1046.0	311.4	58.9	33.0	30.0	868.2	681	2.64	26.3	19	149	89

Surface area	Second moment of area		Radius of gyration		Elastic modulus		Plastic modulus		Buckling parameter	Torsional index	Warping constant	Torsional constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³	u	x	H	J
2.96	325300	9424	35.7	6.07	7204	621	8351	982	0.854	46.8	18.4	291
2.97	376400	11240	36.3	6.27	8269	739	9535	1163	0.861	41.3	22.1	422
2.98	406100	12270	36.6	6.36	8879	805	10230	1266	0.864	38.6	24.2	515
2.99	436300	13300	36.8	6.42	9501	871	10940	1371	0.866	36.2	26.4	626
3.00	470700	14460	36.9	6.47	10200	943	11770	1487	0.866	33.9	28.8	767
3.01	504200	15600	37.0	6.51	10880	1014	12570	1601	0.867	31.9	31.2	926
3.03	549000	17130	37.1	6.55	11780	1107	13650	1754	0.866	29.7	34.5	1164
3.04	625200	19480	37.7	6.66	13260	1266	15340	2000	0.872	26.4	39.7	1648
3.06	696900	21980	37.9	6.72	14660	1416	17030	2247	0.872	24.1	45.2	2199
3.09	777000	24640	38.1	6.78	16170	1575	18870	2510	0.871	22.2	51.3	2891
3.11	884100	28580	38.3	6.89	18210	1810	21350	2895	0.872	19.9	60.0	4091
3.14	981300	32080	38.5	6.95	20010	2013	23590	3235	0.871	18.4	68.1	5337
3.17	110100	36460	38.8	7.05	22180	2266	26260	3654	0.871	16.8	78.5	7127
3.42	625800	39160	37.8	9.46	13730	1871	15480	2890	0.883	30.1	75.8	1193
3.43	671700	42240	38.0	9.53	14660	2014	16550	3111	0.884	28.3	82.2	1442
3.44	719600	45440	38.2	9.59	15630	2161	17670	3341	0.885	26.7	88.9	1734
3.45	785700	49870	38.4	9.68	16940	2366	19170	3658	0.887	24.8	98.3	2188
3.47	846900	53940	38.6	9.73	18150	2550	20600	3948	0.887	23.3	107	2671
3.48	936400	59170	38.8	9.76	19880	2801	22640	4344	0.887	21.4	119	3500
3.50	1031000	65450	38.9	9.81	21700	3081	24820	4790	0.886	19.8	132	4508
3.53	1144000	73000	39.2	9.89	23840	3415	27390	5320	0.887	18.2	149	5890
3.06	408000	9553	38.0	5.81	8409	637	9807	1020	0.850	45.7	21.5	390
3.08	481300	11760	39.0	6.09	9821	784	11350	1245	0.861	39.9	26.8	582
3.10	554000	14010	40.0	6.36	11190	934	12830	1470	0.873	35.0	32.2	835
3.11	644200	16240	40.1	6.37	12880	1083	14850	1713	0.872	30.7	37.7	1264
3.13	723100	18470	40.3	6.44	14350	1223	16590	1941	0.872	27.9	43.3	1718
3.14	807700	20500	40.2	6.40	15900	1353	18540	2168	0.868	25.5	48.4	2330
3.17	909900	23450	40.4	6.49	17740	1536	20760	2469	0.868	23.1	56.0	3185
3.19	1021000	26730	40.6	6.57	19720	1733	23200	2800	0.867	21.1	64.4	4299
3.22	1131000	29970	40.8	6.63	21620	1925	25570	3126	0.866	19.5	73.0	5576

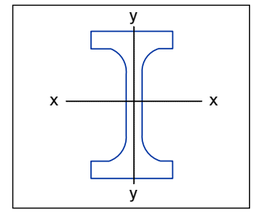
2.4 Metric Beams & Columns (MB & MC)



Metric Units

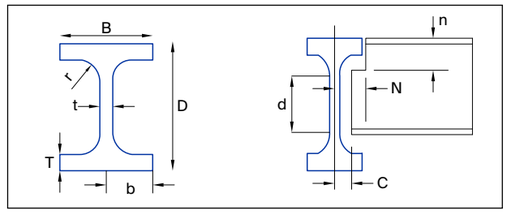
Designation		Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre			Flange	Web				Flange	Web	End clearance	Notch	
		D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
100x50	9.30	100	50	7	5	8	70	11.8	3.57	14.0	5	33	15
100x100	14.8	100	100	7	5	10	66	19.2	7.14	13.2	5	58	17
	16.9	100	100	8	6	8	68	21.6	6.25	11.3	5	57	16
	17.2	100	100	8	6	10	64	21.9	6.25	10.7	5	57	18
125x60	13.2	125	60	8	6	9	91	16.8	3.75	15.2	5	37	17
125x125	23.6	125	125	9	6.5	8	91	30.0	6.94	14.0	5	69	17
	23.8	125	125	9	6.5	10	87	30.3	6.94	13.4	5	69	19
150x75	14.0	150	75	7	5	8	120	17.8	5.36	24.0	5	45	15
150x100	20.7	148	100	9	6	8	114	26.3	5.56	19.0	5	57	17
	21.1	148	100	9	6	11	108	26.8	5.56	18.0	5	57	20
150x150	31.1	150	150	10	7	8	114	39.6	7.50	16.3	6	82	18
	31.5	150	150	10	7	11	108	40.1	7.50	15.4	6	82	21
	37.4	154	151	12	8	11	108	47.7	6.29	13.5	6	82	23
175x90	18.0	175	90	8	5	8	143	22.9	5.63	28.6	5	53	16
	18.1	175	90	8	5	9	141	23.0	5.63	28.2	5	53	17
175x125	23.3	169	125	8	5.5	12	129	29.7	7.81	23.5	5	70	20
175x175	32.8	171	174	9	6	12	129	41.7	9.67	21.5	5	94	21
	40.2	175	175	11	7.5	12	129	51.2	7.95	17.2	6	94	23
	40.4	175	175	11	7.5	13	127	51.4	7.95	16.9	6	94	24
200x100	17.8	198	99	7	4.5	8	168	22.7	7.07	37.3	4	57	15
	18.2	198	99	7	4.5	11	162	23.2	7.07	36.0	4	57	18
	20.9	200	100	8	5.5	8	168	26.7	6.25	30.5	5	57	16
	21.3	200	100	8	5.5	11	162	27.2	6.25	29.5	5	57	19
200x150	29.9	194	150	9	6	8	160	38.1	8.33	26.7	5	82	17
	30.6	194	150	9	6	13	150	39.0	8.33	25.0	5	82	22
	36.9	198	151	11	7	13	150	47.0	6.86	21.4	6	82	24
200x200	41.4	196	199	10	6.5	13	150	52.7	9.95	23.1	5	106	23
	49.9	200	200	12	8	13	150	63.5	8.33	18.8	6	106	25
	57.8	204	201	14	9	13	150	73.6	7.18	16.7	7	106	27
	65.7	208	202	16	10	13	150	83.7	6.31	15.0	7	106	29

Dimensions	JIS G 3192
Specification	EN10025/ ASTM A36/ ASTM A572/ JIS G 3101
Size Range	MB: 100mm x 50mm to 900mm x 300mm MC: 100mm x 100mm to 400mm x 400mm

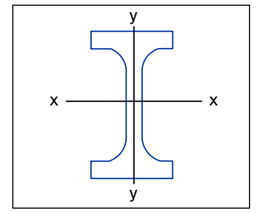


Surface Area per Metre	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³	u	x	H	J
0.376	188	14.8	3.98	1.12	37.5	5.93	44.1	9.5	0.876	12.7	0.000	2.03
0.570	344	117.0	4.20	2.47	69.0	23.0	78.0	36.0	0.843	12.3	0.003	3.53
0.574	378	134	4.18	2.49	75.6	26.7	86.4	41.0	0.836	11.1	0.003	4.73
0.571	383	134	4.18	2.47	76.5	26.8	87.6	41.2	0.837	10.7	0.003	5.17
0.463	413	29.2	4.95	1.32	66.1	9.75	77.6	15.7	0.872	14.0	0.001	3.75
0.723	840	293	5.29	3.13	134	46.9	152	71.7	0.839	12.8	0.010	7.91
0.720	847	294	5.29	3.11	136	47.0	154	71.9	0.840	12.5	0.010	8.43
0.576	666	49.5	6.11	1.67	88.8	13.2	102	20.8	0.876	20.4	0.003	2.81
0.670	1003	150	6.17	2.39	135	30.1	154	46.4	0.880	15.8	0.007	6.56
0.665	1021	151	6.17	2.37	138	30.1	157	46.7	0.881	15.0	0.007	7.37
0.872	1623	563	6.40	3.77	216	75.1	243	114	0.841	14.1	0.028	12.5
0.867	1641	563	6.39	3.75	219	75.1	246	115	0.842	13.7	0.028	13.5
0.877	2018	690	6.51	3.80	262	91.4	298	140	0.845	11.8	0.035	22.2
0.686	1205	97.5	7.26	2.06	138	21.7	156	33.6	0.885	21.8	0.007	4.31
0.685	1214	97.6	7.26	2.06	139	21.7	157	33.7	0.885	21.4	0.007	4.50
0.806	1529	261	7.18	2.97	181	41.8	202	64.3	0.879	19.1	0.017	6.76
1.01	2304	791	7.43	4.35	269	90.9	298	138	0.845	17.5	0.052	11.5
1.01	2884	984	7.50	4.38	330	112	369	171	0.844	14.7	0.066	20.4
1.01	2895	984	7.50	4.37	331	112	370	172	0.845	14.5	0.066	21.0
0.769	1543	113	8.25	2.24	156	22.9	175	35.5	0.883	28.4	0.010	3.30
0.764	1582	114	8.26	2.22	160	23.0	180	35.7	0.884	26.5	0.010	3.86
0.775	1806	134	8.23	2.24	181	26.7	205	41.6	0.878	24.9	0.012	5.08
0.770	1844	134	8.24	2.22	184	26.8	209	41.9	0.880	23.6	0.012	5.77
0.962	2625	507	8.30	3.65	271	67.6	301	103	0.876	21.2	0.043	9.32
0.954	2690	507	8.30	3.61	277	67.7	309	104	0.878	19.8	0.043	10.9
0.964	3331	633	8.42	3.67	336	83.8	377	128	0.879	16.9	0.055	18.5
1.15	3846	1315	8.54	5.00	392	132	433	201	0.847	18.2	0.114	17.6
1.16	4716	1602	8.62	5.02	472	160	525	244	0.846	15.5	0.142	29.8
1.17	5603	1897	8.73	5.08	549	189	617	287	0.847	13.6	0.171	45.8
1.18	6531	2201	8.83	5.13	628	218	710	332	0.848	12.2	0.203	66.7

2.4
Metric Beams & Columns (MB & MC)



Dimensions	JIS G 3192
Specification	EN10025/ ASTM A36/ ASTM A572/ JIS G 3101
Size Range	MB: 100mm x 50mm to 900mm x 300mm MC: 100mm x 100mm to 400mm x 400mm

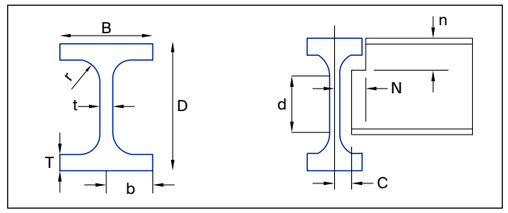


Metric Units

Designation		Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre			Flange	Web				Flange	Web	End clearance	Notch	
		D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
250x125	25.1	248	124	8	5	8	216	32.0	7.75	43.2	5	70	16
	25.7	248	124	8	5	12	208	32.7	7.75	41.6	5	70	20
	29.0	250	125	9	6	8	216	37.0	6.94	36.0	5	70	17
	29.6	250	125	9	6	12	208	37.7	6.94	34.7	5	70	21
250x175	43.6	244	175	11	7	13	196	55.5	7.95	28.0	6	94	24
	44.1	244	175	11	7	16	190	56.2	7.95	27.1	6	94	27
	51.6	248	176	13	8	16	190	65.7	6.77	23.8	6	94	29
	59.1	252	177	15	9	16	190	75.3	5.90	21.1	7	94	31
250x250	66.5	248	249	13	8	16	190	84.7	9.58	23.8	6	131	29
	71.8	250	250	14	9	13	196	91.4	8.93	21.8	7	131	27
	72.4	250	250	14	9	16	190	92.2	8.93	21.1	7	131	30
	81.6	250	255	14	14	13	196	104	9.11	14.0	9	131	27
	98.1	260	253	19	12	16	190	125	6.66	15.8	8	131	35
300x150	25.0	294	148	6	4.5	16	250	32.6	12.3	55.6	4	82	22
	32.0	298	149	8	5.5	13	256	40.8	9.31	46.5	5	82	21
	36.7	300	150	9	6.5	13	256	46.8	8.33	39.4	5	82	22
	41.4	304	150	11	6.5	13	256	52.8	6.82	39.4	5	82	24
	46.2	306	151	12	7.5	13	256	58.8	6.29	34.1	6	82	25
	69.0	318	154	18	11	13	256	87.9	4.28	23.3	8	82	31
300x200	48.3	290	199	10	7	18	234	61.5	9.95	33.4	6	106	28
	55.8	294	200	12	8	13	244	71.1	8.33	30.5	6	106	25
	56.8	294	200	12	8	18	234	72.4	8.33	29.3	6	106	30
	65.4	298	201	14	9	18	234	83.4	7.18	26.0	7	106	32
	77.3	304	202	17	10	18	234	98.5	5.94	23.4	7	106	35
300x300	87.0	298	299	14	9	18	234	111	10.7	26.0	7	155	32
	93.0	300	300	15	10	13	244	118	10.0	24.4	7	155	28
	94.0	300	300	15	10	18	234	120	10.0	23.4	7	155	33
	105	300	305	15	15	13	244	133	10.2	16.3	10	155	28
	106	304	301	17	11	18	234	135	8.85	21.3	8	155	35
	125	310	303	20	13	18	234	159	7.58	18.0	9	155	38
	130	310	305	20	15	18	234	165	7.63	15.6	10	155	38
	147	312	310	21	20	18	234	187	7.38	11.7	12	155	39

Surface Area per Metre	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
0.968	3450	255	10.4	2.82	278	41.1	312	63.2	0.884	32.0	0.037	5.78
0.961	3537	255	10.4	2.79	285	41.1	319	63.6	0.886	29.9	0.037	6.74
0.974	3965	294	10.4	2.82	317	47.0	358	72.7	0.880	28.4	0.043	8.51
0.967	4052	294	10.4	2.79	324	47.0	366	73.1	0.881	26.9	0.043	9.68
1.15	6037	984	10.4	4.21	495	112	550	172	0.883	21.3	0.134	21.2
1.15	6122	985	10.4	4.19	502	113	558	173	0.884	20.5	0.134	23.2
1.16	7308	1184	10.5	4.24	589	135	660	207	0.884	18.0	0.163	36.1
1.17	8541	1390	10.7	4.30	678	157	764	241	0.885	16.0	0.195	53.2
1.45	9931	3348	10.8	6.29	801	269	883	408	0.850	17.9	0.462	46.7
1.46	10750	3648	10.8	6.32	860	292	953	443	0.847	17.1	0.508	55.8
1.45	10830	3649	10.8	6.29	867	292	960	444	0.848	16.7	0.508	58.7
1.47	11400	3876	10.5	6.11	912	304	1031	467	0.826	15.7	0.540	75.4
1.48	15340	5134	11.1	6.41	1180	406	1330	618	0.849	12.8	0.745	141
1.14	4940	326	12.3	3.16	336	44.0	375	68.4	0.878	39.7	0.068	5.51
1.16	6318	442	12.4	3.29	424	59.4	475	91.8	0.880	35.6	0.093	8.65
1.16	7210	508	12.4	3.29	481	67.7	542	105	0.876	32.0	0.107	12.4
1.17	8578	620	12.7	3.43	564	82.7	633	128	0.888	27.8	0.133	18.8
1.18	9514	691	12.7	3.43	622	91.5	702	142	0.884	25.6	0.149	24.9
1.21	14820	1100	13.0	3.54	932	143	1070	223	0.884	17.9	0.248	79.0
1.33	9431	1317	12.4	4.63	650	132	721	203	0.882	26.0	0.258	22.8
1.35	11110	1602	12.5	4.75	756	160	842	245	0.882	24.0	0.319	31.4
1.34	11340	1604	12.5	4.71	771	160	859	247	0.883	22.7	0.319	35.8
1.35	13310	1900	12.6	4.77	893	189	1000	291	0.884	20.1	0.383	53.4
1.37	16280	2341	12.9	4.88	1071	232	1204	356	0.888	17.2	0.482	87.8
1.74	18850	6242	13.0	7.51	1265	418	1389	634	0.850	20.0	1.26	71.3
1.76	20190	6753	13.1	7.55	1346	450	1484	683	0.847	19.4	1.37	82.1
1.75	20410	6756	13.1	7.51	1361	450	1501	684	0.848	18.8	1.37	88.1
1.77	21310	7103	12.6	7.30	1421	466	1596	714	0.828	17.9	1.44	109
1.76	23380	7733	13.2	7.57	1538	514	1705	781	0.849	16.9	1.59	125
1.78	28130	9282	13.3	7.64	1815	613	2031	932	0.849	14.6	1.95	200
1.78	28630	9470	13.2	7.57	1847	621	2079	949	0.842	14.4	1.99	215
1.79	31370	10450	13.0	7.48	2011	674	2295	1040	0.830	13.0	2.21	298

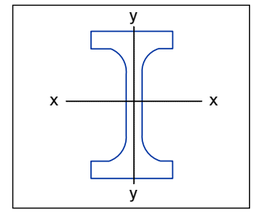
2.4
Metric Beams & Columns (MB & MC)



Metric Units

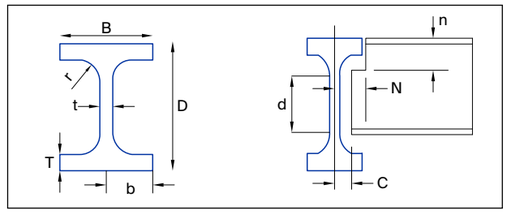
Designation		Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
				Flange	Web				Flange	Web	End clearance	Notch	
Size	Mass per Metre	D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
350x175	41.2	346	174	9	6	13	302	52.5	9.67	50.3	5	94	22
	41.4	346	174	9	6	14	300	52.7	9.67	50.0	5	94	23
	49.4	350	175	11	7	13	302	62.9	7.95	43.1	6	94	24
	49.6	350	175	11	7	14	300	63.1	7.95	42.9	6	94	25
	57.8	354	176	13	8	14	300	73.7	6.77	37.5	6	94	27
	66.2	358	177	15	9	14	300	84.3	5.90	33.3	7	94	29
	71.8	360	178	16	10	14	300	91.4	5.56	30.0	7	94	30
	79.7	364	177	18	11	14	300	101	4.92	27.3	8	93	32
350x250	69.2	336	249	12	8	20	272	88.2	10.4	34.0	6	131	32
	78.1	340	250	14	9	13	286	100	8.93	31.8	7	131	27
	79.7	340	250	14	9	20	272	102	8.93	30.2	7	131	34
	94.2	346	251	17	10	20	272	120	7.38	27.2	7	131	37
	108	350	253	19	12	20	272	137	6.66	22.7	8	131	39
350x350	113	344	348	16	10	13	286	144	10.9	28.6	7	179	29
	115	344	354	16	10	20	272	148	11.1	27.2	7	182	36
	135	350	350	19	12	13	286	172	9.21	23.8	8	179	32
	137	350	350	19	12	20	272	174	9.21	22.7	8	179	39
	159	356	352	22	14	20	272	202	8.00	19.4	9	179	42
	181	362	354	25	16	20	272	230	7.08	17.0	10	179	45
400x200	56.1	396	199	11	7	13	348	71.4	9.05	49.7	6	106	24
	56.6	396	199	11	7	16	342	72.2	9.05	48.9	6	106	27
	65.4	400	200	13	8	13	348	83.4	7.69	43.5	6	106	26
	66.0	400	200	13	8	16	342	84.1	7.69	42.8	6	106	29
	75.5	404	201	15	9	16	342	96.2	6.70	38.0	7	106	31
	88.2	410	202	18	10	16	342	112	5.61	34.2	7	106	34
	140	430	208	28	16	16	342	179	3.71	21.4	10	106	44
	187	446	214	36	22	16	342	239	2.97	15.5	13	106	52
400x300	94.3	386	299	14	9	22	314	120	10.7	34.9	7	155	36
	105	390	300	16	10	13	332	133	9.38	33.2	7	155	29
	107	390	300	16	10	22	314	136	9.38	31.4	7	155	38
	127	396	302	19	12	22	314	162	7.95	26.2	8	155	41
	144	402	303	22	13	22	314	184	6.89	24.2	9	155	44

Dimensions	JIS G 3192
Specification	EN10025/ ASTM A36/ ASTM A572/ JIS G 3101
Size Range	MB: 100mm x 50mm to 900mm x 300mm MC: 100mm x 100mm to 400mm x 400mm

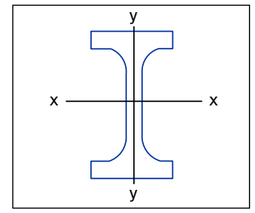


Surface Area per Metre	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
1.35	11040	792	14.5	3.88	638	91.0	712	140	0.881	38.1	0.225	13.2
1.35	11100	792	14.5	3.88	641	91.0	716	140	0.882	37.5	0.225	13.6
1.36	13500	984	14.6	3.96	771	113	864	173	0.883	32.2	0.283	22.4
1.36	13560	985	14.7	3.95	775	113	868	174	0.884	31.8	0.283	23.0
1.37	16100	1184	14.8	4.01	909	135	1022	208	0.885	27.6	0.344	36.1
1.38	18710	1390	14.9	4.06	1045	157	1180	243	0.886	24.4	0.409	53.6
1.39	20240	1508	14.9	4.06	1124	169	1276	263	0.883	22.9	0.446	66.3
1.39	22760	1669	15.0	4.06	1250	189	1425	293	0.884	20.6	0.499	91.3
1.62	18510	3093	14.5	5.92	1102	248	1215	380	0.881	25.8	0.812	44.6
1.64	21230	3649	14.6	6.05	1249	292	1382	445	0.881	24.2	0.969	58.0
1.63	21680	3652	14.6	6.00	1275	292	1412	447	0.882	22.8	0.970	66.3
1.64	26440	4488	14.8	6.12	1528	358	1699	547	0.886	19.5	1.21	109
1.65	30190	5138	14.8	6.12	1725	406	1935	623	0.882	17.6	1.41	155
2.04	32850	11240	15.1	8.84	1910	646	2092	978	0.848	21.1	3.02	111
2.05	33810	11840	15.1	8.95	1966	669	2153	1014	0.846	20.4	3.18	123
2.05	39847	13583	15.2	8.89	2277	776	2515	1176	0.847	18.0	3.72	186
2.04	40300	13590	15.2	8.84	2303	776	2545	1179	0.849	17.5	3.72	199
2.06	47590	16000	15.3	8.90	2674	909	2979	1382	0.848	15.4	4.46	305
2.07	55190	18500	15.5	8.96	3049	1045	3424	1591	0.848	13.7	5.25	445
1.55	19770	1447	16.6	4.50	999	145	1114	223	0.883	36.8	0.536	25.0
1.55	20020	1448	16.7	4.48	1011	145	1128	224	0.884	35.6	0.536	27.1
1.56	23460	1736	16.8	4.56	1173	174	1313	267	0.885	31.8	0.650	39.6
1.56	23710	1737	16.8	4.54	1185	174	1326	268	0.886	30.9	0.650	42.2
1.57	27490	2035	16.9	4.60	1361	202	1528	312	0.887	27.4	0.770	62.3
1.58	33060	2478	17.2	4.70	1612	245	1815	378	0.891	23.4	0.952	101
1.63	54850	4216	17.5	4.86	2551	405	2941	632	0.887	15.7	1.70	373
1.68	75250	5919	17.8	4.98	3374	553	3968	873	0.883	12.5	2.49	825
1.91	33680	6246	16.7	7.21	1745	418	1918	637	0.880	25.8	2.16	79.9
1.94	37860	7204	16.9	7.35	1942	480	2141	730	0.879	24.5	2.52	100
1.92	38680	7210	16.9	7.28	1983	481	2188	733	0.881	23.2	2.52	114
1.94	46660	8735	17.0	7.35	2357	578	2620	884	0.880	20.0	3.10	185
1.95	54420	10220	17.2	7.45	2707	674	3022	1030	0.883	17.7	3.69	272

2.4
Metric Beams & Columns (MB & MC)



Dimensions	JIS G 3192
Specification	EN10025/ ASTM A36/ ASTM A572/ JIS G 3101
Size Range	MB: 100mm x 50mm to 900mm x 300mm MC: 100mm x 100mm to 400mm x 400mm

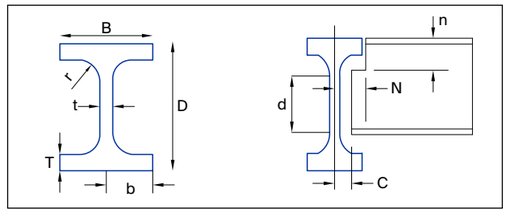


Metric Units

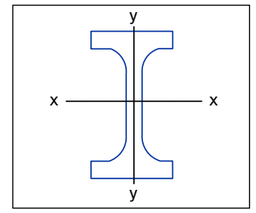
Designation		Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre			Flange	Web				Flange	Web	End clearance	Notch	
		D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
400x400	140	388	402	15	15	22	314	178	13.4	20.9	10	204	37
	147	394	398	18	11	22	314	187	11.1	28.5	8	204	40
	172	400	400	21	13	22	314	219	9.52	24.2	9	204	43
	197	400	408	21	21	22	314	251	9.71	15.0	13	204	43
	200	406	403	24	16	22	314	255	8.40	19.6	10	204	46
	232	414	405	28	18	22	314	295	7.23	17.4	11	204	50
	235	408	412	25	25	22	314	300	8.24	12.6	15	204	47
	283	428	407	35	20	22	314	361	5.81	15.7	12	204	57
	415	458	417	50	30	22	314	529	4.17	10.5	17	204	72
	605	498	432	70	45	22	314	770	3.09	7.0	25	204	92
450x200	65.1	446	199	12	8	13	396	83.0	8.29	49.5	6	106	25
	66.2	446	199	12	8	18	386	84.3	8.29	48.3	6	106	30
	74.9	450	200	14	9	13	396	95.4	7.14	44.0	7	106	27
	76.0	450	200	14	9	18	386	96.8	7.14	42.9	7	106	32
	88.9	456	201	17	10	18	386	113	5.91	38.6	7	106	35
	98.9	460	202	19	11	18	386	126	5.32	35.1	8	106	37
	110	460	205	19	14	18	386	140	5.39	27.6	9	106	37
450x300	106	434	299	15	10	24	356	135	9.97	35.6	7	155	39
	121	440	300	18	11	13	378	154	8.33	34.4	8	155	31
	124	440	300	18	11	24	356	157	8.33	32.4	8	155	42
	145	446	302	21	13	24	356	184	7.19	27.4	9	155	45
500x200	77.9	496	199	14	9	13	442	99	7.11	49.1	7	105	27
	79.5	496	199	14	9	20	428	101	7.11	47.6	7	105	34
	88.2	500	200	16	10	13	442	112	6.25	44.2	7	105	29
	89.6	500	200	16	10	20	428	114	6.25	42.8	7	105	36
	102	506	201	19	11	13	442	129	5.29	40.2	8	105	32
	103	506	201	19	11	20	428	131	5.29	38.9	8	105	39
	117	512	202	22	12	20	428	148	4.59	35.7	8	105	42
500x300	111	482	300	15	11	13	426	141	10.0	38.7	8	155	28
	114	482	300	15	11	26	400	146	10.0	36.4	8	155	41
	125	488	300	18	11	13	426	159	8.33	38.7	8	155	31
	128	488	300	18	11	26	400	164	8.33	36.4	8	155	44
	150	494	302	21	13	26	400	191	7.19	30.8	9	155	47

Surface Area per Metre	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
2.32	48970	16260	16.6	9.55	2524	809	2802	1237	0.830	22.5	5.66	156
2.32	56150	18930	17.3	10.1	2850	951	3118	1441	0.850	20.9	6.69	194
2.34	66620	22420	17.5	10.1	3331	1121	3672	1700	0.850	18.2	8.05	303
2.35	70890	23810	16.8	9.75	3545	1167	3992	1794	0.828	16.7	8.55	415
2.35	78040	26200	17.5	10.1	3844	1300	4280	1977	0.846	16.1	9.56	462
2.37	92770	31030	17.7	10.2	4482	1532	5026	2331	0.848	14.1	11.6	714
2.38	86470	29200	17.0	9.87	4239	1418	4818	2185	0.827	14.2	10.7	694
2.41	119200	39360	18.2	10.4	5570	1934	6311	2941	0.854	11.6	15.2	1317
2.49	187100	60530	18.8	10.7	8172	2903	9540	4436	0.853	8.5	25.2	3885
2.60	297900	94360	19.7	11.1	11960	4369	14460	6724	0.852	6.4	43.2	11060
1.65	28130	1579	18.4	4.36	1262	159	1423	245	0.876	38.4	0.744	33.9
1.64	28700	1581	18.5	4.33	1287	159	1450	247	0.878	36.5	0.744	38.3
1.66	32890	1870	18.6	4.43	1462	187	1652	290	0.878	33.6	0.889	51.6
1.65	33450	1872	18.6	4.40	1487	187	1679	291	0.879	32.2	0.890	56.9
1.67	40400	2308	18.9	4.51	1772	230	2003	356	0.884	27.5	1.11	92.5
1.68	45430	2619	19.0	4.56	1975	259	2240	403	0.885	25.0	1.27	126
1.68	47860	2742	18.5	4.43	2081	268	2399	423	0.869	23.9	1.33	152
2.00	46800	6695	18.6	7.04	2157	448	2384	686	0.883	27.0	2.94	104
2.04	54730	8106	18.9	7.26	2488	540	2757	823	0.885	24.9	3.61	141
2.02	56070	8114	18.9	7.18	2549	541	2825	828	0.887	23.4	3.61	163
2.03	66380	9658	19.0	7.24	2977	640	3323	981	0.886	20.5	4.36	253
1.75	40840	1843	20.3	4.31	1647	185	1869	288	0.874	37.5	1.07	52.5
1.74	41870	1846	20.3	4.27	1688	186	1914	290	0.876	35.2	1.07	60.8
1.76	46810	2138	20.4	4.36	1872	214	2130	333	0.875	33.3	1.25	75.9
1.75	47850	2142	20.5	4.33	1914	214	2175	335	0.877	31.6	1.25	85.9
1.77	55480	2578	20.7	4.47	2193	257	2496	399	0.880	28.7	1.53	119
1.76	56520	2582	20.7	4.43	2234	257	2541	401	0.881	27.5	1.53	132
1.77	65450	3035	21.0	4.52	2557	300	2913	469	0.884	24.3	1.82	193
2.12	58270	6756	20.3	6.92	2418	450	2696	690	0.877	32.4	3.68	93.9
2.10	60370	6768	20.4	6.82	2505	451	2791	695	0.880	29.4	3.69	118
2.13	68860	8106	20.8	7.14	2822	540	3132	825	0.887	28.0	4.48	143
2.11	70960	8118	20.8	7.05	2908	541	3228	830	0.889	25.9	4.48	172
2.13	83810	9663	20.9	7.11	3393	640	3792	984	0.888	22.8	5.40	265

2.4
Metric Beams & Columns (MB & MC)



Dimensions	JIS G 3192
Specification	EN10025/ ASTM A36/ ASTM A572/ JIS G 3101
Size Range	MB: 100mm x 50mm to 900mm x 300mm MC: 100mm x 100mm to 400mm x 400mm



Metric Units

Designation		Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre			Flange	Web				Flange	Web	End clearance	Notch	
		D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
600x200	79.0	592	197	13	8	22	522	101	7.58	65.3	6	105	35
	92.5	596	199	15	10	13	540	118	6.63	54.0	7	105	28
	94.6	596	199	15	10	22	522	120	6.63	52.2	7	105	37
	103	600	200	17	11	13	540	132	5.88	49.1	8	105	30
	106	600	200	17	11	22	522	134	5.88	47.5	8	105	39
	118	606	201	20	12	13	540	150	5.03	45.0	8	105	33
	120	606	201	20	12	22	522	152	5.03	43.5	8	105	42
	134	612	202	23	13	22	522	171	4.39	40.2	9	105	45
600x300	133	582	300	17	12	13	522	169	8.82	43.5	8	154	30
	137	582	300	17	12	28	492	174	8.82	41.0	8	154	45
	147	588	300	20	12	13	522	187	7.50	43.5	8	154	33
	151	588	300	20	12	28	492	192	7.50	41.0	8	154	48
	170	594	302	23	14	13	522	217	6.57	37.3	9	154	36
	175	594	302	23	14	28	492	222	6.57	35.1	9	154	51
	203	602	304	27	16	28	492	259	5.63	30.8	10	154	55
	217	608	304	30	16	28	492	277	5.07	30.8	10	154	58
700x300	166	692	300	20	13	28	596	211	7.50	45.8	9	154	48
	182	700	300	24	13	18	616	232	6.25	47.4	9	154	42
	185	700	300	24	13	28	596	235	6.25	45.8	9	154	52
	215	708	302	28	15	28	596	274	5.39	39.7	10	154	56
800x300	191	792	300	22	14	28	692	243	6.82	49.4	9	153	50
	207	800	300	26	14	18	712	264	5.77	50.9	9	153	44
	210	800	300	26	14	28	692	267	5.77	49.4	9	153	54
	241	808	302	30	16	28	692	308	5.03	43.3	10	153	58
	267	816	303	34	17	28	692	340	4.46	40.7	11	153	62
900x300	210	890	299	23	15	18	808	267	6.50	53.9	10	152	41
	213	890	299	23	15	28	788	271	6.50	52.5	10	152	51
	240	900	300	28	16	18	808	306	5.36	50.5	10	152	46
	243	900	300	28	16	28	788	310	5.36	49.3	10	152	56
	283	912	302	34	18	18	808	360	4.44	44.9	11	152	52
	286	912	302	34	18	28	788	364	4.44	43.8	11	152	62
	304	918	303	37	19	18	808	387	4.09	42.5	12	152	55
	307	918	303	37	19	28	788	391	4.09	41.5	12	152	65

Surface Area per Metre	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
1.92	58240	1665	24.1	4.07	1968	169	2239	265	0.870	45.2	1.40	53.0
1.95	66640	1976	23.8	4.10	2236	199	2576	312	0.861	42.9	1.67	69.2
1.93	68720	1982	23.9	4.06	2306	199	2651	315	0.864	39.7	1.67	82.4
1.96	75560	2274	24.0	4.16	2519	227	2904	358	0.863	38.4	1.93	97.3
1.94	77630	2280	24.0	4.12	2588	228	2979	361	0.865	36.0	1.94	113
1.97	88320	2716	24.3	4.26	2915	270	3357	426	0.868	33.4	2.33	147
1.95	90400	2723	24.3	4.23	2983	271	3432	429	0.870	31.7	2.34	167
1.97	103500	3178	24.6	4.32	3382	315	3893	498	0.873	28.3	2.76	237
2.32	98950	7659	24.2	6.73	3400	511	3822	786	0.876	35.5	6.11	137
2.29	102700	7675	24.3	6.63	3530	512	3963	793	0.879	32.1	6.13	173
2.33	114400	9009	24.7	6.94	3889	601	4348	921	0.886	31.2	7.27	199
2.30	118100	9025	24.8	6.85	4018	602	4489	928	0.888	28.8	7.28	241
2.35	133600	10570	24.8	6.98	4497	700	5057	1077	0.885	27.3	8.62	304
2.32	137300	10590	24.9	6.90	4624	701	5197	1085	0.887	25.5	8.63	356
2.34	162600	12680	25.1	7.00	5401	834	6101	1292	0.888	22.3	10.5	551
2.35	179300	14090	25.4	7.13	5896	927	6653	1431	0.893	20.5	11.8	702
2.51	172400	9030	28.6	6.53	4984	602	5629	936	0.878	34.3	10.2	260
2.54	197500	10820	29.2	6.83	5643	721	6338	1110	0.887	31.4	12.4	343
2.53	201500	10830	29.3	6.78	5757	722	6464	1116	0.889	30.0	12.4	383
2.55	237100	12890	29.4	6.86	6699	854	7559	1323	0.889	26.3	14.9	588
2.71	253600	9936	32.3	6.39	6405	662	7288	1036	0.873	36.8	14.7	341
2.74	286400	11720	33.0	6.67	7159	781	8098	1210	0.882	33.9	17.6	440
2.72	291700	11740	33.0	6.62	7292	782	8243	1216	0.883	32.5	17.6	486
2.74	339200	13820	33.2	6.70	8397	915	9534	1426	0.883	28.7	20.9	726
2.76	383600	15820	33.6	6.82	9402	1044	10680	1625	0.887	25.8	24.2	1003
2.92	338500	10280	35.6	6.20	7608	687	8750	1079	0.863	42.2	19.3	361
2.90	345300	10290	35.7	6.16	7760	688	8913	1085	0.865	40.2	19.3	403
2.94	404500	12630	36.4	6.43	8989	842	10290	1317	0.872	35.9	24.0	580
2.92	411300	12650	36.4	6.39	9139	843	10450	1324	0.873	34.5	24.0	633
2.97	491000	15660	36.9	6.59	10770	1037	12340	1622	0.876	30.1	30.2	979
2.95	497800	15670	37.0	6.56	10920	1038	12500	1629	0.877	29.3	30.2	1050
2.98	535400	17210	37.2	6.67	11660	1136	13380	1778	0.878	27.9	33.4	1234
2.96	542200	17230	37.2	6.63	11810	1137	13540	1785	0.879	27.2	33.4	1316